

switching the orders of integration. To illustrate this idea, we prove the following typical theorem on analyticity of integrals.

Theorem. *Suppose that $h(t, z)$ is a continuous complex-valued function, defined for $a \leq t \leq b$ and $z \in D$. If for each fixed t , $h(t, z)$ is an analytic function of $z \in D$, then*

$$H(z) = \int_a^b h(t, z) dt, \quad z \in D,$$

is analytic on D .

To see this, note first that $H(z)$ is continuous on D . Indeed, if $z_n \rightarrow z$, then $h(t, z_n) \rightarrow h(t, z)$ uniformly for $a \leq t \leq b$, so $H(z_n) \rightarrow H(z)$. Let R be a closed rectangle in D . By Cauchy's theorem, we have

$$\int_{\partial R} h(t, z) dz = 0.$$

Consequently,

$$\int_a^b \int_{\partial R} h(t, z) dz dt = 0.$$

Parameterization of the sides of ∂R converts the inside integral to a sum of four garden-variety integrals of a continuous function, and we can interchange the order of integration. This yields

$$0 = \int_{\partial R} \int_a^b h(t, z) dt dz = \int_{\partial R} H(z) dz.$$

The hypotheses of Morera's theorem are met, and $H(z)$ is analytic.

As another typical application, we prove the following useful result.

Theorem. *Suppose that $f(z)$ is a continuous function on a domain D that is analytic on $D \setminus \mathbb{R}$, that is, on the part of D not lying on the real axis. Then $f(z)$ is analytic on D .*

Let R be a closed rectangle contained in D , with sides parallel to the coordinate axes. To prove the theorem, it suffices to show that $\int_{\partial R} f(z) dz = 0$. There are three cases to consider. If the closed rectangle R does not meet the real axis $\mathbb{R}$, then $f(z)$ is analytic on R , so the integral is zero by Cauchy's theorem. For the second case, suppose that R has one edge that lies on the real axis, say the lower edge of R is an interval $[a, b]$ on the real axis. For $\varepsilon > 0$ small, let R_ε be the closed rectangle in the upper half-plane consisting of $z \in R$ such that $\text{Im } z \geq \varepsilon$. By Cauchy's theorem, $\int_{\partial R_\varepsilon} f(z) dz = 0$. We claim that

$$(6.1) \quad \int_{\partial R_\varepsilon} f(z) dz \rightarrow \int_{\partial R} f(z) dz$$

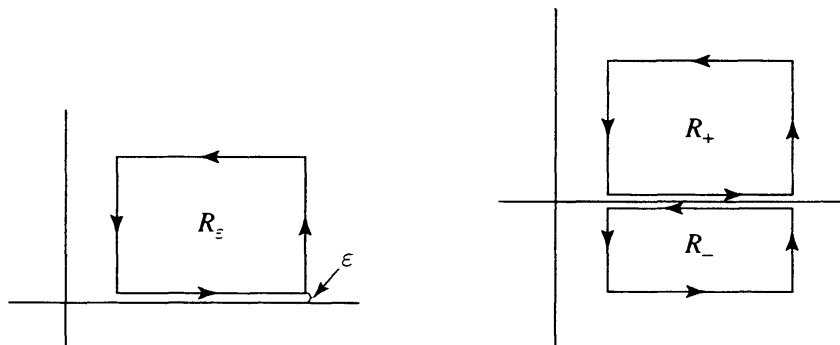
as $\varepsilon \rightarrow 0$. Indeed the integral along the bottom edge of R_ε has the form

$$\int_a^b f(t + i\varepsilon) dt,$$

and since $f(t + i\varepsilon)$ converges to $f(t)$ as $\varepsilon \rightarrow 0$, uniformly for $a \leq t \leq b$, the integrals converge to the integral $\int_a^b f(t)dt$ of $f(z)$ along the bottom edge of R . The top edge of R_ε coincides with the top edge of R , and the vertical sides of R_ε differ from the vertical sides of R only by vertical intervals of length ε , whose contribution to the integral tends to 0 with ε . Thus (6.1) holds, and we conclude in this case that $\int_{\partial R} f(z)dz = 0$. Finally, if the top edge of R is in the upper half-plane and the bottom edge of R is in the lower half-plane, we define R_+ to be the part of R in the closed upper half-plane and R_- to be the part of R in the closed lower half-plane. Then the integrals of $f(z)$ around ∂R_+ and ∂R_- are both zero, by case two. Thus

$$\int_{\partial R} f(z)dz = \int_{\partial R_+} f(z)dz + \int_{\partial R_-} f(z)dz = 0,$$

and the analyticity follows from Morera's theorem.



Exercises for IV.6

1. Let L be a line in the complex plane. Suppose $f(z)$ is a continuous complex-valued function on a domain D that is analytic on $D \setminus L$. Show that $f(z)$ is analytic on D .
2. Let $h(t)$ be a continuous function on the interval $[a, b]$. Show that the Fourier transform

$$H(z) = \int_a^b h(t)e^{-itz} dt$$

is an entire function that satisfies

$$|H(z)| \leq Ce^{A|y|}, \quad z = x + iy \in \mathbb{C},$$

for some constants $A, C > 0$. *Remark.* An entire function satisfying such a growth restriction is called an **entire function of finite type**.

3. Let $h(t)$ be a continuous function on a subinterval $[a, b]$ of $[0, \infty)$. Show that the Fourier transform $H(z)$, defined as above, is bounded in the lower half-plane.
4. Let γ be a smooth curve in the plane $\mathbb{R}^2$, and let D be a domain in the complex plane, and let $P(x, y, \zeta)$ and $Q(x, y, \zeta)$ be continuous complex-valued functions defined for (x, y) on γ and $\zeta \in D$. Suppose that the functions depend analytically on ζ for each fixed (x, y) on γ . Show that

$$G(\zeta) = \int_{\gamma} P(x, y, \zeta) dx + Q(x, y, \zeta) dy$$

is analytic on D .

7. Goursat's Theorem

We have defined $f(z)$ to be analytic on D if the complex derivative $f'(z)$ exists at each point of D and further, $f'(z)$ is a continuous function of z . Goursat's theorem asserts that the requirement that $f'(z)$ be continuous is redundant.

Theorem (Goursat's Theorem). *If $f(z)$ is a complex-valued function on a domain D such that*

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists at each point z_0 of D , then $f(z)$ is analytic on D .

Goursat's theorem is as useless as it is aesthetically pleasing. In applications it has always turned out that if one can show the existence of a complex derivative at each point, then one can see with little more effort that the complex derivative is continuous. Nevertheless, the idea of the proof has proved to be very useful in other contexts.

The proof is based on Morera's theorem. Let R be a closed rectangle in D . We subdivide R into four equal subrectangles. Since the integral of $f(z)$ around ∂R is the sum of the integrals of $f(z)$ around the four subrectangles, there is at least one of the subrectangles, call it R_1 , for which

$$\left| \int_{\partial R_1} f(z) dz \right| \geq \frac{1}{4} \left| \int_{\partial R} f(z) dz \right|.$$

Now subdivide R_1 into four equal subrectangles and repeat the procedure. This yields a nested sequence of rectangles $\{R_n\}$ such that

$$\left| \int_{\partial R_n} f(z) dz \right| \geq \frac{1}{4} \left| \int_{\partial R_{n-1}} f(z) dz \right| \geq \cdots \geq \frac{1}{4^n} \left| \int_{\partial R} f(z) dz \right|.$$

Since the R_n 's are decreasing and have diameters tending to 0, the R_n 's converge to some point $z_0 \in D$. Since $f(z)$ is differentiable at z_0 , we have an estimate of the form

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| \leq \varepsilon_n, \quad z \in R_n,$$

where $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$. Let L be the length of ∂R . Then the length of ∂R_n is $L/2^n$. For z belonging to R_n we have the estimate

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon_n |z - z_0| \leq 2\varepsilon_n L/2^n.$$

From the ML -estimate and Cauchy's theorem, we obtain

$$\begin{aligned} \left| \int_{\partial R_n} f(z) dz \right| &= \left| \int_{\partial R_n} [f(z) - f(z_0) - f'(z_0)(z - z_0)] dz \right| \\ &\leq (2\varepsilon_n L/2^n) \cdot (L/2^n) = 2L^2 \varepsilon_n / 4^n. \end{aligned}$$

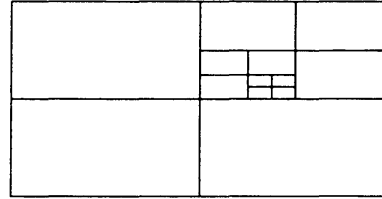
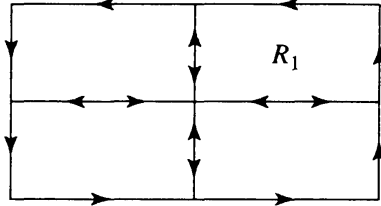
Hence

$$\left| \int_{\partial R} f(z) dz \right| \leq 4^n \left| \int_{\partial R_n} f(z) dz \right| \leq 2L^2 \varepsilon_n.$$

Since $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$, we must have

$$\int_{\partial R} f(z) dz = 0.$$

By Morera's theorem, $f(z)$ is analytic.



Exercises for IV.7

1. Find an application for Goursat's theorem in which it is not patently clear by other means that the function in question is analytic.

8. Complex Notation and Pompeiu's Formula

Many results in complex analysis can be expressed very simply in terms of the first-order differential operators

$$\frac{\partial}{\partial z} = \frac{1}{2} \left[\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right], \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left[\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right].$$

We may think of $\frac{\partial f}{\partial z}$ as an average of the derivatives of $f(z)$ in the x and the iy directions,

$$\frac{\partial f}{\partial z} = \frac{1}{2} \left[\frac{\partial f}{\partial x} + \frac{\partial f}{\partial(iy)} \right].$$

When we derived the Cauchy-Riemann equations in Chapter II, we found two expressions for the derivative of an analytic function $f(z)$,

$$f'(z) = \frac{\partial f}{\partial x} \quad \text{and} \quad f'(z) = -i \frac{\partial f}{\partial y} = \frac{\partial f}{\partial(iy)}.$$

If we take the average of these two expressions for $f'(z)$, we obtain

$$(8.1) \quad f'(z) = \frac{\partial f}{\partial z},$$

again provided that $f(z)$ is analytic.

To understand the operator $\partial/\partial\bar{z}$, we write $f = u + iv$ and compute

$$\frac{\partial f}{\partial\bar{z}} = \frac{1}{2} \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right] + \frac{i}{2} \left[\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right].$$

If we equate the real and imaginary parts of the right-hand side to zero, we obtain the Cauchy-Riemann equations for u and v . Thus the equation

$$(8.2) \quad \frac{\partial f}{\partial\bar{z}} = 0$$

is equivalent to the Cauchy-Riemann equations for u and v . Equation (8.2) is referred to as the **complex form of the Cauchy-Riemann equations**. We summarize our observations.

Theorem. *Let $f(z)$ be a continuously differentiable function on a domain D . Then $f(z)$ is analytic if and only if $f(z)$ satisfies the complex form (8.2) of the Cauchy-Riemann equations. If $f(z)$ is analytic, then the derivative of $f(z)$ is given by (8.1).*

We list some rules for operating with $\partial/\partial z$ and $\partial/\partial\bar{z}$. Since these are both first-order differential operators with constant coefficients, they are linear,

$$\frac{\partial}{\partial z}(af + bg) = a \frac{\partial f}{\partial z} + b \frac{\partial g}{\partial z}, \quad \frac{\partial}{\partial\bar{z}}(af + bg) = a \frac{\partial f}{\partial\bar{z}} + b \frac{\partial g}{\partial\bar{z}},$$

and they satisfy the Leibniz rule,

$$\frac{\partial}{\partial z}(fg) = f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z}, \quad \frac{\partial}{\partial\bar{z}}(fg) = f \frac{\partial g}{\partial\bar{z}} + g \frac{\partial f}{\partial\bar{z}}.$$

The z -derivative and the $\bar{z}$ -derivative are related to each other by

$$\frac{\partial f}{\partial\bar{z}} = \overline{\frac{\partial \bar{f}}{\partial z}}, \quad \frac{\partial \bar{f}}{\partial\bar{z}} = \overline{\frac{\partial f}{\partial z}}.$$

These formulae can be verified by writing $f = u + iv$, $\bar{f} = u - iv$, and carrying out the calculations.

The Taylor series expansion of a smooth function $f(z)$ at z_0 , through only the linear terms, is given by

$$f(z) = f(z_0) + \frac{\partial f}{\partial z}(z_0)(z - z_0) + \frac{\partial f}{\partial \bar{z}}(z_0)(\overline{z - z_0}) + \mathcal{O}(|z - z_0|^2),$$

where the “big oh” term is a remainder term bounded by $C|z - z_0|^2$. The complex Taylor expansion can be derived from the usual Taylor expansion by substituting the definitions and calculating. It can also be derived by observing that it suffices to check the formula for the three functions 1, $z - z_0$, and $\overline{z - z_0}$.

As an application of Taylor’s formula, we derive the complex form of the formula for the tangent vector to a curve. Let $f(z)$ be a smooth function, and let $\gamma(t)$ be a smooth curve terminating at $\gamma(0) = z_0$. From the linear Taylor series approximation we have

$$f(\gamma(t)) - f(\gamma(0)) = \frac{\partial f}{\partial z}(z_0)(\gamma(t) - z_0) + \frac{\partial f}{\partial \bar{z}}(z_0)(\overline{\gamma(t) - z_0}) + \mathcal{O}(|\gamma(t) - z_0|^2)$$

Dividing by t and taking the limit as $t \rightarrow 0$, we obtain

$$(8.3) \quad (f \circ \gamma)'(0) = \frac{\partial f}{\partial z}(z_0)\gamma'(0) + \frac{\partial f}{\partial \bar{z}}(z_0)\overline{\gamma'(0)}.$$

We use this formula to show that conformal maps are analytic.

Theorem. *Let $f(z)$ be a continuously differentiable function on a domain D . Suppose that the gradient of $f(z)$ does not vanish at any point of D , and that $f(z)$ is conformal. Then $f(z)$ is analytic on D , and $f'(z) \neq 0$ on D .*

To prove this, fix a point $z_0 \in D$, and consider the straight line $\gamma(t) = z_0 + te^{i\theta}$, $0 \leq t \leq \varepsilon$, terminating at z_0 with tangent $e^{i\theta}$. By (8.3), the image curve $f \circ \gamma$ has tangent at $f(z_0)$ given by

$$(f \circ \gamma)'(0) = \frac{\partial f}{\partial z}(z_0)e^{i\theta} + \frac{\partial f}{\partial \bar{z}}(z_0)e^{-i\theta}.$$

The condition on the gradient guarantees that this is not identically zero. In order for $f(z)$ to preserve angles at z_0 , the difference in the arguments of $(f \circ \gamma)'(0)$ and $\gamma'(0)$ must be constant, independent of θ . Hence the argument of

$$\frac{(f \circ \gamma)'(0)}{\gamma'(0)} = \frac{\partial f}{\partial z}(z_0) + \frac{\partial f}{\partial \bar{z}}(z_0)e^{-2i\theta}$$

must be independent of θ . However, this occurs only when $\partial f / \partial \bar{z} = 0$. Hence $f(z)$ is analytic on D . The gradient condition implies that $f'(z) = \partial f / \partial z \neq 0$, and the theorem is established. Note that this theorem could

as well be proved using x and y derivatives rather than z and $\bar{z}$ derivatives. However, the complex notation makes the proof more transparent.

For the case of complex line integrals, Green's theorem yields the following formula, which can be regarded as an extension of Cauchy's theorem to arbitrary smooth functions.

Theorem. *If D is a bounded domain in the complex plane with piecewise smooth boundary, and if $g(z)$ is a smooth function on $D \cup \partial D$, then*

$$(8.4) \quad \int_{\partial D} g(z) dz = 2i \iint_D \frac{\partial g}{\partial \bar{z}} dx dy.$$

To see this we replace dz by $dx + i dy$ and apply Green's formula:

$$\int_{\partial D} g dx + \int_{\partial D} i g dy = \iint_D \left(i \frac{\partial g}{\partial x} - \frac{\partial g}{\partial y} \right) dx dy = 2i \iint_D \frac{\partial g}{\partial \bar{z}} dx dy.$$

Note that if $g(z)$ is analytic on D , then $\partial g / \partial \bar{z} = 0$ on D , so the integral over D vanishes, and we obtain Cauchy's theorem.

Cauchy's integral formula can also be extended to apply to arbitrary smooth functions.

Theorem (Pompeiu's Formula). *Suppose D is a bounded domain with piecewise smooth boundary. If $g(z)$ is a smooth complex-valued function on $D \cup \partial D$, then*

$$(8.5) \quad g(w) = \frac{1}{2\pi i} \int_{\partial D} \frac{g(z)}{z - w} dz - \frac{1}{\pi} \iint_D \frac{\partial g}{\partial \bar{z}} \frac{1}{z - w} dx dy, \quad w \in D.$$

Pompeiu's formula is established by the same argument as was used in Section 4 to establish Cauchy's integral formula, except that now the correction term appears in the calculation. Let D_ε be the domain obtained from D by punching out a disk centered at w of radius ε . We apply the complex version (8.4) of Green's theorem to the function $g(z)/(z - w)$. For this, note that

$$\frac{\partial}{\partial \bar{z}} \left(\frac{g(z)}{z - w} \right) = \frac{\partial g}{\partial \bar{z}} \frac{1}{z - w} + g \frac{\partial}{\partial \bar{z}} \left(\frac{1}{z - w} \right) = \frac{\partial g}{\partial \bar{z}} \frac{1}{z - w}, \quad z \in D_\varepsilon,$$

so that by (8.4),

$$(8.6) \quad \int_{\partial D_\varepsilon} \frac{g(z)}{z - w} dz = 2i \iint_{D_\varepsilon} \frac{\partial g}{\partial \bar{z}} \frac{1}{z - w} dx dy.$$

The singularity of $1/(z - w)$ at $z = w$ is absolutely integrable:

$$\iint_{|z-w| \leq 1} \frac{1}{|z - w|} dx dy = \int_0^{2\pi} \int_0^1 \frac{1}{r} \cdot r dr d\theta = 2\pi < \infty.$$

Hence the area integral in (8.6) over D_ε tends to the (improper) area integral over D as $\varepsilon \rightarrow 0$. The boundary integral in (8.6) has the form

$$\int_{\partial D_\varepsilon} \frac{g(z)}{z-w} dz = \int_{\partial D} \frac{g(z)}{z-w} dz - \int_{|z-w|=\varepsilon} \frac{g(z)}{z-w} dz.$$

If we parametrize the circle $|z-w| = \varepsilon$, we obtain for the integral on the right

$$\int_{|z-w|=\varepsilon} \frac{g(z)}{z-w} dz = i \int_0^{2\pi} g(w + \varepsilon e^{i\theta}) d\theta.$$

This tends to $2\pi i g(w)$ as $\varepsilon \rightarrow 0$, since $g(z)$ is continuous at w . Thus if we let $\varepsilon \rightarrow 0$ in (8.6), we obtain

$$\int_{\partial D} \frac{g(z)}{z-w} dz - 2\pi i g(w) = 2i \iint_D \frac{\partial g}{\partial \bar{z}} \frac{1}{z-w} dx dy,$$

which is equivalent to (8.5).

The formula (8.5) is also known as the **Cauchy-Green formula**, since it is proved using Green's theorem. The formula can be regarded as Cauchy's integral formula with a correction term added to account for the fact that $g(z)$ may not be analytic. If $g(z)$ is analytic on D , it reduces to the Cauchy integral formula (4.1) for $g(z)$.

Exercises for IV.8

1. Show from the definition that

$$\frac{\partial}{\partial z} z = 1, \quad \frac{\partial}{\partial \bar{z}} z = 0, \quad \frac{\partial}{\partial z} \bar{z} = 0, \quad \frac{\partial}{\partial \bar{z}} \bar{z} = 1.$$

2. Compute $\frac{\partial}{\partial \bar{z}}(az^2 + bz\bar{z} + c\bar{z}^2)$. Use the result to determine where $az^2 + bz\bar{z} + c\bar{z}^2$ is complex-differentiable and where it is analytic. (See Problem II.2.3.)
3. Show that the Jacobian of a smooth function f is given by

$$J_f = \left| \frac{\partial f}{\partial z} \right|^2 - \left| \frac{\partial f}{\partial \bar{z}} \right|^2.$$

4. Show that

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4 \frac{\partial^2}{\partial z \partial \bar{z}}.$$

Deduce the following, for a smooth complex-valued function h .

- (a) h is harmonic if and only if $\partial^2 h / \partial z \partial \bar{z} = 0$.
- (b) h is harmonic if and only if $\partial h / \partial z$ is analytic.
- (c) h is harmonic if and only if $\partial h / \partial \bar{z}$ is conjugate-analytic.

- (d) If h is harmonic, then any m th order partial derivative of h is a linear combination of $\partial^m h / \partial z^m$ and $\partial^m h / \partial \bar{z}^m$.

5. With $d\bar{z} = dx - i dy$, show for a smooth function $f(z)$ that

$$df = \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z}.$$

6. Show that if D is a domain with smooth boundary, and if $f(z)$ and $g(z)$ are analytic on $D \cup \partial D$, then

$$\int_{\partial D} f(z) \overline{g(z)} dz = 2i \iint_D f(z) \overline{g'(z)} dx dy.$$

Compare this formula with Exercise 1.4.

7. Show that the Taylor series expansion at $z_0 = 0$ of a smooth function $f(z)$, through the quadratic terms, is given by

$$\begin{aligned} f(z) = & f(0) + \frac{\partial f}{\partial z}(0)z + \frac{\partial f}{\partial \bar{z}}(0)\bar{z} \\ & + \frac{1}{2} \left[\frac{\partial^2 f}{\partial z^2}(0)z^2 + 2 \frac{\partial^2 f}{\partial z \partial \bar{z}}(0)|z|^2 + \frac{\partial^2 f}{\partial \bar{z}^2}(0)\bar{z}^2 \right] + \mathcal{O}(|z|^3). \end{aligned}$$

8. Establish the following version of the chain rule for smooth complex-valued functions $w = w(z)$ and $h = h(w)$:

$$\begin{aligned} \frac{\partial}{\partial \bar{z}}(h \circ w) &= \frac{\partial h}{\partial w} \frac{\partial w}{\partial \bar{z}} + \frac{\partial h}{\partial \bar{w}} \frac{\partial \bar{w}}{\partial \bar{z}}, \\ \frac{\partial}{\partial z}(h \circ w) &= \frac{\partial h}{\partial w} \frac{\partial w}{\partial z} + \frac{\partial h}{\partial \bar{w}} \frac{\partial \bar{w}}{\partial z}. \end{aligned}$$

9. Show with the aid of the preceding exercise that if both $h(w)$ and $w(z)$ are analytic, then $(h \circ w)(z)$ is analytic, and $(h \circ w)'(z) = h'(w(z))w'(z)$.
10. Let $g(z)$ be a continuously differentiable function on the complex plane that is zero outside of some compact set. Show that

$$g(w) = -\frac{1}{\pi} \iint_{\mathbb{C}} \frac{\partial g}{\partial \bar{z}} \frac{1}{z-w} dx dy, \quad w \in \mathbb{C}.$$

Remark. If we integrate this formally by parts, we obtain

$$g(w) = \frac{1}{\pi} \iint_{\mathbb{C}} g(z) \frac{\partial}{\partial \bar{z}} \left(\frac{1}{z-w} \right) dx dy.$$

Thus the “distribution derivative” of $1/(\pi(z-w))$ with respect to z is the point mass at w (“Dirac delta-function”), in the sense that it is equal to 0 away from w , and it is infinite at w in such a way that its integral (total mass) is equal to 1.

V

Power Series

In this chapter we show that the analytic functions are exactly the functions that can be expanded in a convergent power series about any point. Since power series can be treated very much as polynomials, this provides a powerful tool for dealing with analytic functions. In Sections 1 and 2 we review infinite series and series of functions. Sections 3 through 6 contain the basic material on power series. In Section 7 we use power series to show that the zeros of an analytic function are isolated. This leads to the uniqueness principle for analytic functions. Section 8 contains a formal definition of analytic continuation, which can be omitted at first reading.

1. Infinite Series

In this section we review some basic material on infinite series of complex numbers. In the next section we provide some background material on sequences and series of functions. The reader may wish to skip to Section 3 and refer to the background sections only when necessary.

A series $\sum_{k=0}^{\infty} a_k$ of complex numbers is said to **converge to S** if the sequence of partial sums $\{S_k\}$, defined by $S_k = a_0 + \cdots + a_k$, converges to S . For notation, the sum S of the series is denoted by $\sum_{k=0}^{\infty} a_k$, or simply by $\sum a_k$.

Any statement concerning series can be reinterpreted as a statement about sequences, by phrasing the statement in terms of the sequence of partial sums of the series. For instance, we know that if $s_n \rightarrow s$ and $t_n \rightarrow t$, then $s_n + t_n \rightarrow s + t$. If we apply this statement to the partial sums of series, we conclude that if $\sum a_k = A$ and $\sum b_k = B$, then $\sum (a_k + b_k) = A + B$. Similarly, since $s_n \rightarrow s$ implies $cs_n \rightarrow cs$, we deduce that if $\sum a_k = A$, then $\sum ca_k = cA$. Thus taking limits of sequences is a linear operation, and this implies that summing series is also a linear operation.

If the r_k 's are positive real numbers, then the partial sums $S_k = r_0 + \cdots + r_k$ form a monotone increasing sequence. Since a monotone sequence of real numbers converges if and only if it is bounded, we see that a series of

positive numbers converges if and only if the partial sums are bounded. If $0 \leq a_k \leq r_k$, and if the partial sums of $\sum r_k$ are bounded, then the partial sums of $\sum a_k$ are bounded by the same bound. This observation leads to the following convergence test for series with positive terms.

Theorem (Comparison Test). *If $0 \leq a_k \leq r_k$, and if $\sum r_k$ converges, then $\sum a_k$ converges, and $\sum a_k \leq \sum r_k$.*

The terms of a series can be recovered from the partial sums, by $a_k = S_k - S_{k-1}$. Suppose the series $\sum a_k$ converges to S . Then S_k converges to S as $k \rightarrow \infty$, and also S_{k-1} converges to S as $k \rightarrow \infty$. Hence $a_k \rightarrow 0$ as $k \rightarrow \infty$. This simple necessary condition provides a useful screening test for convergence.

Theorem. *If $\sum a_k$ converges, then $a_k \rightarrow 0$ as $k \rightarrow \infty$.*

Example. The most important series for us will be the geometric series $\sum_{k=0}^{\infty} z^k$. The k th partial sum of the geometric series is given by

$$S_k = 1 + z + z^2 + \cdots + z^{k-1} + z^k = \frac{1 - z^{k+1}}{1 - z}, \quad z \neq 1.$$

To see this, we multiply and divide the sum by $1 - z$, and we note that the numerator telescopes:

$$(1 - z)S_k = 1 - z + z - z^2 + \cdots + z^{k-1} - z^k + z^k - z^{k+1} = 1 - z^{k+1}.$$

If $|z| < 1$, then $z^{k+1} \rightarrow 0$ and $S_k \rightarrow 1/(1 - z)$ as $k \rightarrow \infty$. Hence

$$\sum_{k=0}^{\infty} z^k = \frac{1}{1 - z}, \quad |z| < 1.$$

On the other hand, if $|z| \geq 1$, then the k th term z^k does not converge to 0, so that the series does not converge.

The series $\sum a_k$ is said to **converge absolutely** if $\sum |a_k|$ converges. Thus for a series of positive terms, convergence and absolute convergence are the same. There are convergent series that are not absolutely convergent. (See Exercises 4 and 5.) However, every absolutely convergent series is convergent.

Theorem. *If $\sum a_k$ converges absolutely, then $\sum a_k$ converges, and*

$$(1.1) \quad \left| \sum_{k=0}^{\infty} a_k \right| \leq \sum_{k=0}^{\infty} |a_k|.$$

The proof is easy, modulo a little trick, which is to express $\operatorname{Re} a_k$ as a difference $\operatorname{Re} a_k = (\operatorname{Re} a_k + |a_k|) - |a_k|$. Since $|\operatorname{Re} a_k| \leq |a_k|$, we have

$0 \leq \operatorname{Re} a_k + |a_k| \leq 2|a_k|$. Hence $\sum(\operatorname{Re} a_k + |a_k|)$ is a series of non-negative real numbers. Since its partial sums are bounded, by $2\sum|a_k|$, the series converges. Now $\sum \operatorname{Re} a_k$ is the difference of two convergent series, $\sum(\operatorname{Re} a_k + |a_k|)$ and $\sum|a_k|$, and so $\sum \operatorname{Re} a_k$ converges. Similarly, $\sum \operatorname{Im} a_k$ converges, and consequently $\sum a_k$ converges. For each N we have $|\sum_{k=0}^N a_k| \leq \sum_{k=0}^N |a_k|$. Letting $N \rightarrow \infty$, we obtain the estimate (1.1).

Example. The geometric series converges absolutely when $|z| < 1$, and

$$\left| \frac{1}{1-z} \right| = \left| \sum_{k=0}^{\infty} z^k \right| \leq \sum_{k=0}^{\infty} |z|^k = \frac{1}{1-|z|}, \quad |z| < 1.$$

This leads also to a useful estimate for the difference between the partial sums of the geometric series and the full sum. From

$$(1.2) \quad \frac{1}{1-z} - \sum_{k=0}^n z^k = \sum_{k=n+1}^{\infty} z^k = z^{n+1} \sum_{j=0}^{\infty} z^j = \frac{z^{n+1}}{1-z}$$

we obtain

$$\left| \frac{1}{1-z} - \sum_{k=0}^n z^k \right| \leq \frac{|z|^{n+1}}{1-|z|}, \quad |z| < 1.$$

Exercises for V.1

1. (Harmonic Series) Show that

$$\sum_{k=1}^n \frac{1}{k} \geq \log n.$$

Deduce that the series $\sum \frac{1}{k}$ does not converge. *Hint.* Use the estimate

$$\frac{1}{k} \geq \int_k^{k+1} \frac{1}{x} dx.$$

2. Show that if $p < 1$, then the series $\sum_{k=1}^{\infty} 1/k^p$ diverges. *Hint.* Use Exercise 1 and the comparison test.
3. Show that if $p > 1$, then the series $\sum_{k=1}^{\infty} 1/k^p$ converges to S , where

$$\left| S - \sum_{k=1}^n \frac{1}{k^p} \right| < \frac{1}{(p-1)n^{p-1}}.$$

Hint. Use the estimate $\frac{1}{k^p} < \int_{k-1}^k \frac{dx}{x^p}$.

4. Show that the series

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

converges. *Hint.* Show that the partial sums of the series satisfy $S_1 < S_3 < S_5 < \cdots$ and $S_2 > S_4 > S_6 > \cdots$.

5. Show that the series

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots$$

converges to $3S/2$, where S is the sum of the series in Exercise 4. (It turns out that $S = \log 2$.) *Hint.* Organize the terms in the series in Exercise 4 in groups of four, and relate it to the groups of three in the above series.

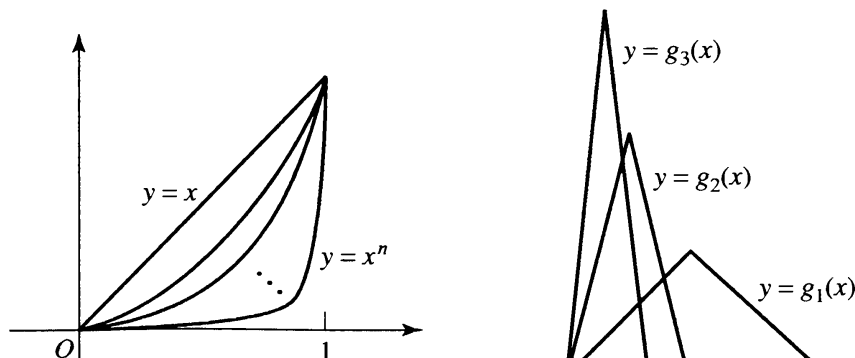
6. Show that $\sum \frac{1}{k \log k}$ diverges while $\sum \frac{1}{k(\log k)^2}$ converges.

7. Show that the series $\sum a_k$ converges if and only if $\sum_{k=m}^{k=n} a_k$ tends to 0 as $m, n \rightarrow \infty$. *Remark.* This is the **Cauchy criterion for series**.

2. Sequences and Series of Functions

Let $\{f_j\}$ be a sequence of complex-valued functions defined on some set E . We say that the sequence $\{f_j\}$ **converges pointwise** on E if for each point $x \in E$ the sequence of complex numbers $\{f_j(x)\}$ converges. The limit $f(x)$ of $\{f_j(x)\}$ is then a complex-valued function on E .

Example. The sequence of functions $f_j(x) = x^j$, $0 \leq x \leq 1$, converges pointwise on the unit interval to the function $f(x)$ defined on the unit interval by $f(x) = 0$ for $0 \leq x < 1$ and $f(1) = 1$. Note that the pointwise limit of a sequence of continuous functions need not be continuous.

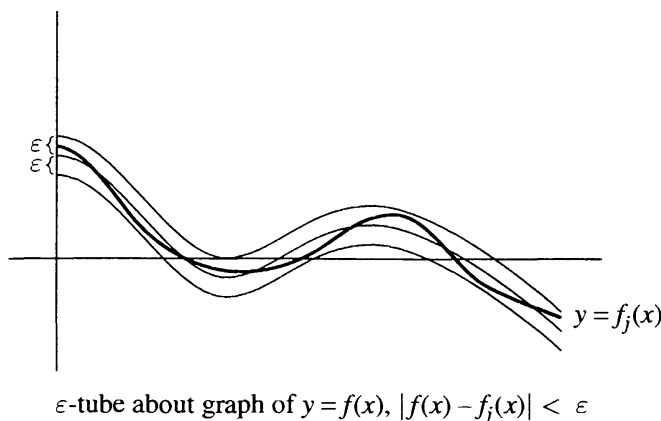


Example. We define a sequence of “tent functions” g_j on the unit interval by $g_j(x) = j^2x$ for $0 \leq x \leq 1/j$, $g_j(x) = 2j - j^2x$ for $1/j \leq x \leq 2/j$, and $g_j(x) = 0$ for $2/j \leq x \leq 1$. The height of the j th tent is j , and the width of the base is $2/j$, so that the area under the tent is $\int_0^1 g_j(x) dx = 1$. On the other hand, the sequence of functions $g_j(x)$ converges pointwise to 0 on the unit interval, and the integral of the pointwise limit is 0.

To guarantee that the limit of the integrals of a sequence of functions is the integral of the limit, we must require that the functions converge in some stronger sense than pointwise. Toward this goal we introduce the notion of uniform convergence. We say that the sequence $\{f_j\}$ of functions on E **converges uniformly** to f on E if $|f_j(x) - f(x)| \leq \varepsilon_j$ for all $x \in E$, where $\varepsilon_j \rightarrow 0$ as $j \rightarrow \infty$. We may regard ε_j as a worst-case estimator for the difference $f_j(x) - f(x)$, and usually we take ε_j to be the supremum (maximum) of $|f_j(x) - f(x)|$ over $x \in E$,

$$\varepsilon_j = \sup_{x \in E} |f_j(x) - f(x)|.$$

Note that if $\{f_j\}$ converges uniformly to f on E , then it converges pointwise to f on E .



In the two examples above, the sequences do not converge uniformly. In the first example the worst-case estimator is $\varepsilon_j = \max_{0 \leq x \leq 1} x^j = 1$, which does not converge to 0. In the example of the sequence of tent functions, the worst-case estimator is worse. It is the height of the tent, which is $\varepsilon_j = j \rightarrow +\infty$.

It turns out that with uniform instead of pointwise convergence, the two theorems we desire are valid: (1) a uniform limit of continuous functions is continuous, and (2) an integral of a uniform limit is the limit of the integrals. We state the theorems more precisely.

Theorem. Let $\{f_j\}$ be a sequence of complex-valued functions defined on a subset E of the complex plane. If each f_j is continuous on E , and if $\{f_j\}$ converges uniformly to f on E , then f is continuous on E .

Theorem. Let γ be a piecewise smooth curve in the complex plane. If $\{f_j\}$ is a sequence of continuous complex-valued functions on γ , and if $\{f_j\}$ converges uniformly to f on γ , then $\int_{\gamma} f_j(z)dz$ converges to $\int_{\gamma} f(z)dz$.

The first theorem above on the continuity of a uniform limit has a standard formal proof, which we omit. The second theorem above on the limit of the integrals is an easy consequence of the ML -estimate. Indeed, suppose $\{f_j\}$ converges uniformly to f on γ . Let ε_j be the worst-case estimator for $f_j - f$ on γ , so that $|f_j - f| \leq \varepsilon_j$ on γ , and let L be the length of γ . Then the ML -estimate gives

$$\left| \int_{\gamma} f_j(z)dz - \int_{\gamma} f(z)dz \right| \leq \varepsilon_j L,$$

and this tends to 0, since the f_j 's converge uniformly to f . Hence $\int f_j dz$ tends to $\int f dz$.

Now we turn to series of functions. Let $\sum g_j(x)$ be a series of complex-valued functions defined on a set E . The partial sums of the series are the functions

$$S_n(x) = \sum_{k=0}^n g_k(x) = g_0(x) + g_1(x) + \cdots + g_n(x).$$

We say that the series **converges pointwise** on E if the sequence of partial sums converges pointwise on E , and the series **converges uniformly** on E if the sequence of partial sums converges uniformly on E . The following criterion for uniform convergence of a series of functions is extremely useful. In fact, it is the only test for uniform convergence of series that we will ever need.

Theorem (Weierstrass M -Test). Suppose $M_k \geq 0$ and $\sum M_k$ converges. If $g_k(x)$ are complex-valued functions on a set E such that $|g_k(x)| \leq M_k$ for all $x \in E$, then $\sum g_k(x)$ converges uniformly on E .

The proof is straightforward. For each fixed x , the estimate for $g_k(x)$ shows that the series $\sum g_k(x)$ is absolutely convergent, and $\sum |g_k(x)| \leq \sum M_k$. By the theorem in Section 1, the series $\sum g_k(x)$ converges to some complex number $g(x)$, and by (1.1), $|g(x)| \leq \sum |g_k(x)| \leq \sum M_k$. The same estimate, applied to the tail of the series, shows that

$$\left| g(x) - S_n(x) \right| = \left| \sum_{k=n+1}^{\infty} g_k(x) \right| \leq \sum_{k=n+1}^{\infty} M_k.$$

If we set $\varepsilon_n = \sum_{k=n+1}^{\infty} M_k$, then $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$, and the estimate shows that the partial sums $S_n(x)$ converge uniformly on E to $g(x)$.

Example. For the geometric series

$$\sum_{k=0}^{\infty} z^k = \frac{1}{1-z}, \quad |z| < 1,$$

we have from (1.2) that

$$\left| \frac{1}{1-z} - S_n(z) \right| = \left| \frac{z^{n+1}}{1-z} \right|,$$

which tends to $+\infty$ as $z \rightarrow 1$. Hence the partial sums do not converge uniformly for $|z| < 1$. However, suppose we fix a radius $r < 1$. Define $M_k = r^k$. Then $\sum M_k$ converges, and $|z^k| \leq M_k$ for $|z| \leq r$. By the Weierstrass M -test, $\sum z^k$ converges uniformly for $|z| \leq r$. Thus the geometric series converges uniformly on each disk $\{|z| \leq r\}$, for each $r < 1$, but it does not converge uniformly on the disk $\{|z| < 1\}$.

Now we return to analytic functions. We begin by proving that a uniform limit of analytic functions is analytic.

Theorem. *If $\{f_k(z)\}$ is a sequence of analytic functions on a domain D that converges uniformly to $f(z)$ on D , then $f(z)$ is analytic on D .*

This can be proved using the Cauchy integral formula. An easier way to see it is to apply Morera's theorem. Since analytic functions are continuous, and the limit of a uniformly convergent sequence of continuous functions is continuous, $f(z)$ is continuous. Let E be a closed rectangle contained in D . By Cauchy's theorem, $\int_{\partial E} f_k(z) dz = 0$ for each k . From the theorem above we obtain in the limit that $\int_{\partial E} f(z) dz = 0$. By Morera's theorem, $f(z)$ is analytic.

Theorem. *Suppose that $f_k(z)$ is analytic for $|z - z_0| \leq R$, and suppose that the sequence $\{f_k(z)\}$ converges uniformly to $f(z)$ for $|z - z_0| \leq R$. Then for each $r < R$ and for each $m \geq 1$, the sequence of m th derivatives $\{f_k^{(m)}(z)\}$ converges uniformly to $f^{(m)}(z)$ for $|z - z_0| \leq r$.*

To prove this, suppose $\varepsilon_k \rightarrow 0$ are such that $|f_k(z) - f(z)| < \varepsilon_k$ for $|z - z_0| < R$. Fix s such that $r < s < R$. The Cauchy integral formula for the m th derivative of $f_k(z) - f(z)$ on the disk $|z - z_0| \leq s$ yields

$$f_k^{(m)}(z) - f^{(m)}(z) = \frac{m!}{2\pi i} \int_{|z-z_0|=s} \frac{f_k(\zeta) - f(\zeta)}{(\zeta - z)^{m+1}} d\zeta, \quad |z - z_0| \leq r.$$

If $|\zeta - z_0| = s$ and $|z - z_0| \leq r$, then $|\zeta - z| \geq s - r$, and so

$$\left| \frac{f_k(\zeta) - f(\zeta)}{(\zeta - z)^{m+1}} \right| \leq \frac{\varepsilon_k}{(s - r)^{k+1}}.$$

From the ML -estimate, we obtain

$$\left| f_k^{(m)}(z) - f^{(m)}(z) \right| \leq \frac{m!}{2\pi} \frac{\varepsilon_k}{(s-r)^{k+1}} 2\pi s = \rho_k, \quad |z - z_0| \leq r.$$

Since $\rho_k \rightarrow 0$ as $k \rightarrow \infty$, we obtain uniform convergence of the m th derivatives for $|z - z_0| \leq r$.

We say that a sequence $\{f_k(z)\}$ of analytic functions on a domain D **converges normally** to the analytic function $f(z)$ on D if it converges uniformly to $f(z)$ on each closed disk contained in D . It is easy to see that this occurs if and only if $\{f_k(z)\}$ converges to $f(z)$ uniformly on each bounded subset E of D at a strictly positive distance from the boundary of D . (See Exercises 4 and 5.) Since any closed disk contained in D can be dilated to a larger disk contained in D , we can apply the preceding theorem, and we obtain the following.

Theorem. Suppose that $\{f_k(z)\}$ is a sequence of analytic functions on a domain D that converges normally on D to the analytic function $f(z)$. Then for each $m \geq 1$, the sequence of m th derivatives $\{f_k^{(m)}(z)\}$ converges normally to $f^{(m)}(z)$ on D .

Exercises for V.2

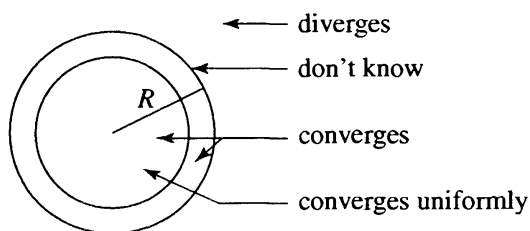
1. Show that $f_k(x) = x^k/(k + x^{2k})$ converges uniformly to 0 on $[0, \infty)$.
Hint. Determine the worst-case estimator ε_k by calculus.
2. Show that $g_k(x) = x^k/(1 + x^k)$ converges pointwise on $[0, \infty)$ but not uniformly. What is the limit function? On which subsets of $[0, \infty)$ does the sequence converge uniformly?
3. Show that $f_k(z) = z^k/k$ converges uniformly for $|z| < 1$. Show that $f'_k(z)$ does not converge uniformly for $|z| < 1$. What *can* be said about the uniform convergence of $f'_k(z)$?
4. Show that $\sum \frac{1}{k^2} \frac{x^k}{1 + x^{2k}}$ converges uniformly for $-\infty < x < +\infty$.
5. For which real numbers x does $\sum \frac{1}{k} \frac{x^k}{1 + x^{2k}}$ converge?
6. Show that for each $\varepsilon > 0$, the series $\sum \frac{1}{k} \frac{x^k}{1 + x^{2k}}$ converges uniformly for $x \geq 1 + \varepsilon$.
7. Let a_n be a bounded sequence of complex numbers. Show that for each $\varepsilon > 0$, the series $\sum_{n=1}^{\infty} a_n n^{-z}$ converges uniformly for $\operatorname{Re} z \geq 1 + \varepsilon$. Here we choose the principal branch of n^{-z} .

8. Show that $\sum \frac{z^k}{k^2}$ converges uniformly for $|z| < 1$.
9. Show that $\sum \frac{z^k}{k}$ does not converge uniformly for $|z| < 1$.
10. Show that if a sequence of functions $\{f_k(x)\}$ converges uniformly on E_j for $1 \leq j \leq n$, then the sequence converges uniformly on the union $E = E_1 \cup E_2 \cup \cdots \cup E_n$.
11. Suppose that E is a bounded subset of a domain $D \subset \mathbb{C}$ at a positive distance from the boundary of D , that is, there is $\delta > 0$ such that $|z - w| \geq \delta$ for all $z \in D$ and $w \in \mathbb{C} \setminus D$. Show that E can be covered by a finite number of closed disks contained in D . *Hint.* Consider all closed disks with centers at points $(m + ni)\delta/10$ and radius $\delta/10$ that meet E .
12. Let $f(z)$ be analytic on a domain D , and suppose $|f(z)| \leq M$ for all $z \in D$. Show that for each $\delta > 0$ and $m \geq 1$, $|f^{(m)}(z)| \leq m!M/\delta^m$ for all $z \in D$ whose distance from ∂D is at least δ . Use this to show that if $\{f_k(z)\}$ is a sequence of analytic functions on D that converges uniformly to $f(z)$ on D , then for each m the derivatives $f_k^{(m)}(z)$ converge uniformly to $f^{(m)}(z)$ on each subset of D at a positive distance from ∂D .

3. Power Series

A **power series** (centered at z_0) is a series of the form $\sum_{k=0}^{\infty} a_k(z - z_0)^k$. By making a change of variable $w = z - z_0$, we can always reduce to the case of power series centered at $z = 0$. The main result on convergence of power series is the following.

Theorem. Let $\sum a_k z^k$ be a power series. Then there is R , $0 \leq R \leq +\infty$, such that $\sum a_k z^k$ converges absolutely if $|z| < R$, and $\sum a_k z^k$ does not converge if $|z| > R$. For each fixed r satisfying $r < R$, the series $\sum a_k z^k$ converges uniformly for $|z| \leq r$.



We call R the **radius of convergence** of the series $\sum a_k z^k$. The radius of convergence depends only on the tail of the series. If we alter a finite

number of coefficients of the series, the radius of convergence remains the same.

For the general case of a power series $\sum a_k(z - z_0)^k$, the domain of convergence is a disk $|z - z_0| < R$. The series diverges if $|z - z_0| > R$, and anything can happen when $|z - z_0| = R$.

For the proof of the theorem, note first that if the sequence $|a_k|r^k$ is bounded for some value $r = r_0$, then it is bounded for all r satisfying $0 \leq r < r_0$. We define R , $0 \leq R \leq +\infty$, to be the supremum of the r 's such that $|a_k|r^k$ is bounded. Thus $|a_k|r^k$ is bounded if $r < R$, while if $r > R$, then there is a sequence of terms with $|a_{k_j}|r^{k_j} \rightarrow +\infty$. In the borderline case $r = R$, anything can happen. The sequence $|a_k|R^k$ might be bounded and it might not.

If $|z| > R$, then the terms $a_k z^k$ do not tend to 0, so that the series does not converge. On the other hand, suppose $r < R$. Choose s such that $r < s < R$. Then the sequence $|a_k|s^k$ is bounded, say $|a_k|s^k \leq C$ for $k \geq 0$. If $|z| \leq r$, then

$$|a_k z^k| \leq |a_k| r^k = |a_k| s^k \left(\frac{r}{s}\right)^k \leq C \left(\frac{r}{s}\right)^k.$$

Set $M_k = C(r/s)^k$. Since $\sum M_k$ converges, the Weierstrass M -test applies, and the series $\sum a_k z^k$ converges uniformly for $|z| \leq r$, and also absolutely for each z . This proves the theorem.

Example. The geometric series $\sum z^k$ has radius of convergence $R = 1$. The series does not converge on the boundary circle $|z| = 1$, since the terms do not tend to 0.

Example. The power series $\sum z^k/k^2$ converges uniformly for $|z| \leq 1$. This follows from the Weierstrass M -test, with majorants $M_k = 1/k^2$. On the other hand, if $r > 1$, then $r^k/k^2 \rightarrow \infty$ as $k \rightarrow \infty$. Thus the series does not converge for $|z| > 1$, and the radius of convergence of the series is $R = 1$.

Example. The series

$$(3.1) \quad \sum_{k=0}^{\infty} \frac{(-1)^k}{2^k} z^{2k} = 1 - \frac{z^2}{2} + \frac{z^4}{2^2} - \frac{z^6}{2^3} + \cdots$$

becomes a geometric series if we set $w = -z^2/2$,

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k} z^{2k} = \sum_{k=0}^{\infty} w^k.$$

The series converges precisely when $|w| < 1$, that is, when $|z^2| < 2$. The radius of convergence is thus $R = \sqrt{2}$. The series converges to $1/(1-w) = 2/(2-z^2)$.

Example. The series $\sum k^k z^k$ has radius of convergence $R = 0$. It converges only for $z = 0$, since $k^k r^k \rightarrow +\infty$ for all $r > 0$.

Example. The series $\sum k^{-k} z^k$ has radius of convergence $R = +\infty$. It converges for all z .

The partial sums of a power series are polynomials in z , and in particular they are analytic functions. From the convergence theorem of Section 2, we obtain the following.

Theorem. Suppose $\sum a_k z^k$ is a power series with radius of convergence $R > 0$. Then the function

$$f(z) = \sum_{k=0}^{\infty} a_k z^k, \quad |z| < R,$$

is analytic. The derivatives of $f(z)$ are obtained by differentiating the series term by term,

$$f'(z) = \sum_{k=1}^{\infty} k a_k z^{k-1}, \quad f''(z) = \sum_{k=2}^{\infty} k(k-1) a_k z^{k-2}, \quad |z| < R,$$

and similarly for the higher-order derivatives. The coefficients of the series are given by

$$(3.2) \quad a_k = \frac{1}{k!} f^{(k)}(0), \quad k \geq 0.$$

The formula for the coefficient a_k is obtained by differentiating k times the series for $f(z)$ and plugging in $z = 0$.

Example. By differentiating the representation of $1/(1-z)$ as a geometric series, we obtain a power series representation of $1/(1-z)^2$,

$$\frac{1}{(1-z)^2} = \sum_{k=1}^{\infty} k z^{k-1} = \sum_{m=0}^{\infty} (m+1) z^m, \quad |z| < 1.$$

On account of the uniform convergence of power series on subdisks of radius strictly smaller than R , a power series can be integrated term by term. Thus if $\sum a_k z^k$ has radius of convergence R , then

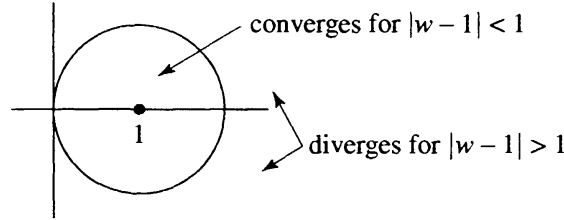
$$\int_0^z \sum (a_k \zeta^k) d\zeta = \sum a_k \int_0^z \zeta^k d\zeta = \sum \frac{a_k}{k+1} z^{k+1}, \quad |z| < R.$$

Example. If we integrate the geometric series term by term, we obtain

$$\begin{aligned} -\operatorname{Log}(1-z) &= \int_0^z \frac{d\zeta}{1-\zeta} = \int_0^z \sum_{k=0}^{\infty} \zeta^k d\zeta \\ &= \sum_{k=0}^{\infty} \frac{z^{k+1}}{k+1} = z + \frac{z^2}{2} + \frac{z^3}{3} + \frac{z^4}{4} + \cdots, \quad |z| < 1. \end{aligned}$$

Making the substitution $w = 1 - z$, we obtain a series expansion for $\operatorname{Log} w$ centered at $w = 1$,

$$\begin{aligned} \operatorname{Log} w &= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (w-1)^k \\ &= (w-1) - \frac{(w-1)^2}{2} + \frac{(w-1)^3}{3} - \frac{(w-1)^4}{4} + \cdots, \quad |w-1| < 1. \end{aligned}$$



Now we turn to two formulae for determining the radius of convergence of a power series from its coefficients. The first of these is based on the **ratio test**. It is especially convenient for determining the radius of convergence of many series that arise as solutions of linear differential equations.

Theorem. If $|a_k/a_{k+1}|$ has a limit as $k \rightarrow \infty$, either finite or $+\infty$, then the limit is the radius of convergence R of $\sum a_k z^k$,

$$R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right|.$$

To see this, let $L = \lim |a_k/a_{k+1}|$. If $r < L$, then $|a_k/a_{k+1}| > r$ eventually, say for $k \geq N$. Then $|a_k| > r|a_{k+1}|$ for $k \geq N$, and

$$|a_N| r^N \geq |a_{N+1}| r^{N+1} \geq |a_{N+2}| r^{N+2} \geq \cdots.$$

Hence the sequence $|a_k| r^k$ is bounded. From the definition of R we have $r \leq R$, and since $r < L$ is arbitrary, we also have $L \leq R$.

Suppose next that $s > L$. Then $|a_k/a_{k+1}| < s$ eventually, say for $k \geq N$. Then $|a_k| < s|a_{k+1}|$ for $k \geq N$, and

$$|a_N| s^N \leq |a_{N+1}| s^{N+1} \leq |a_{N+2}| s^{N+2} \leq \cdots.$$

Hence the terms $a_k z^k$ do not converge to 0 for $|z| \geq s$, so that the series does not converge, and $s \geq R$. Since $s > L$ is arbitrary, we also have $L \geq R$. We conclude that $L = R$.

Example. For the series $\sum k z^k$, the ratio test gives

$$\left| \frac{a_k}{a_{k+1}} \right| = \frac{k}{k+1} \rightarrow 1.$$

Hence the radius of convergence is $R = 1$.

Example. For the series $\sum \frac{z^k}{k!}$, the ratio test gives

$$\left| \frac{a_k}{a_{k+1}} \right| = \frac{(k+1)!}{k!} = k+1 \rightarrow +\infty.$$

Hence the radius of convergence is $R = +\infty$.

The second formula is based on the **root test**.

Theorem. If $\sqrt[k]{|a_k|}$ has a limit as $k \rightarrow \infty$, either finite or $+\infty$, then the radius of convergence of $\sum a_k z^k$ is given by

$$(3.3) \quad R = \frac{1}{\lim \sqrt[k]{|a_k|}}.$$

If $r > 1/\lim \sqrt[k]{|a_k|}$, then $\sqrt[k]{|a_k|} r > 1$ eventually, so that $|a_k| r^k > 1$ eventually, the terms of the series $\sum a_k z^k$ do not converge to 0 for $|z| = r$, and $r \geq R$. On the other hand, if $r < 1/\lim \sqrt[k]{|a_k|}$, then $\sqrt[k]{|a_k|} r < 1$ eventually, so that $|a_k| r^k < 1$ eventually, the sequence $|a_k| r^k < 1$ is bounded, and from the definition of R we have $r \leq R$. It follows that (3.3) holds.

Example. For the series $\sum k z^k$, the root test gives

$$R = 1/\lim \sqrt[k]{k} = 1.$$

There is a more general form of the formula (3.3), called the **Cauchy-Hadamard formula**, that gives the radius of convergence for *any* power series in terms of a lim sup. Recall (Section II.1) that the lim sup of a sequence $\{s_n\}$ is characterized as the number S , $-\infty \leq S \leq +\infty$, with the property that if $t > S$, then only finitely many terms of the sequence satisfy $s_n > t$, while if $t < S$, then infinitely many terms of the sequence satisfy $s_n > t$. If the sequence s_n has a limit, then the lim sup of the sequence coincides with the limit. However, every sequence has a lim sup. The Cauchy-Hadamard formula is obtained simply by replacing the limit in (3.3) by a lim sup,

$$(3.4) \quad R = \frac{1}{\limsup \sqrt[k]{|a_k|}}.$$

The proof is identical to the proof given above, except that the characterizing property of the lim sup is used.

Example. We return to the series (3.1) treated earlier. This can be expressed as a power series $\sum a_k z^k$, where $a_k = (-1)^{k/2}/2^{k/2}$ if k is even, and $a_k = 0$ if k is odd. Thus $\sqrt[k]{|a_k|}$ is $1/\sqrt{2}$ if k is even and 0 if k is odd. The lim sup of this sequence is $1/\sqrt{2}$. By the Cauchy-Hadamard formula, the radius of convergence of the series is $\sqrt{2}$, as before.

Exercises for V.3

1. Find the radius of convergence of the following power series:

$$\begin{array}{lll} \text{(a)} \sum_{k=0}^{\infty} 2^k z^k & \text{(d)} \sum_{k=0}^{\infty} \frac{3^k z^k}{4^k + 5^k} & \text{(g)} \sum_{k=1}^{\infty} \frac{k^k}{1 + 2^k k^k} z^k \\ \text{(b)} \sum_{k=0}^{\infty} \frac{k}{6} z^k & \text{(e)} \sum_{k=1}^{\infty} \frac{2^k z^{2k}}{k^2 + k} & \text{(h)} \sum_{k=3}^{\infty} (\log k)^{k/2} z^k \\ \text{(c)} \sum_{k=1}^{\infty} k^2 z^k & \text{(f)} \sum_{k=1}^{\infty} \frac{z^{2k}}{4^k k^k} & \text{(i)} \sum_{k=1}^{\infty} \frac{k! z^k}{k^k} \end{array}$$

2. Determine for which z the following series converge.

$$\begin{array}{lll} \text{(a)} \sum_{k=1}^{\infty} (z-1)^k & \text{(c)} \sum_{m=0}^{\infty} 2^m (z-2)^m & \text{(e)} \sum_{n=1}^{\infty} n^n (z-3)^n \\ \text{(b)} \sum_{k=10}^{\infty} \frac{(z-i)^k}{k!} & \text{(d)} \sum_{m=1}^{\infty} \frac{(z+i)^m}{m^2} & \text{(f)} \sum_{n=3}^{\infty} \frac{2^n}{n^2} (z-2-i)^n \end{array}$$

3. Find the radius of convergence of the following series.

$$\begin{array}{ll} \text{(a)} \sum_{n=0}^{\infty} z^{3^n} = z + z^3 + z^9 + z^{27} + z^{81} + \dots, \\ \text{(b)} \sum_{p \text{ prime}} z^p = z^2 + z^3 + z^5 + z^7 + z^{11} + \dots \end{array}$$

4. Show that the function defined by $f(z) = \sum z^{n!}$ is analytic on the open unit disk $\{|z| < 1\}$. Show that $|f(r\lambda)| \rightarrow +\infty$ as $r \rightarrow 1$ whenever λ is a root of unity. *Remark.* Thus $f(z)$ does not extend analytically to any larger open set than the open unit disk.

5. What functions are represented by the following power series?

$$\text{(a)} \sum_{k=1}^{\infty} k z^k, \quad \text{(b)} \sum_{k=1}^{\infty} k^2 z^k.$$

6. Show the series $\sum a_k z^k$, the differentiated series $\sum k a_k z^{k-1}$, and the integrated series $\sum \frac{a_k}{k+1} z^{k+1}$ all have the same radius of convergence.
7. Consider the series

$$\sum_{k=0}^{\infty} (2 + (-1)^k)^k z^k.$$

Use the Cauchy-Hadamard formula to find the radius of convergence of the series. What happens when the ratio test is applied? Evaluate explicitly the sum of the series.

8. Write out a proof of the Cauchy-Hadamard formula (3.4).

4. Power Series Expansion of an Analytic Function

We have seen that power series expansions $\sum a_k (z - z_0)^k$ are analytic inside the disk of convergence $\{|z - z_0| < R\}$. It is an important and far-reaching fact that conversely, any function analytic on a disk can be expanded in a power series that converges on the disk.

Theorem. Suppose that $f(z)$ is analytic for $|z - z_0| < \rho$. Then $f(z)$ is represented by the power series

$$(4.1) \quad f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k, \quad |z - z_0| < \rho,$$

where

$$(4.2) \quad a_k = \frac{f^{(k)}(z_0)}{k!}, \quad k \geq 0,$$

and where the power series has radius of convergence $R \geq \rho$. For any fixed r , $0 < r < \rho$, we have

$$(4.3) \quad a_k = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta, \quad k \geq 0.$$

Further, if $|f(z)| \leq M$ for $|z - z_0| = r$, then

$$(4.4) \quad |a_k| \leq \frac{M}{r^k}, \quad k \geq 0.$$

The proof amounts to expanding the integrand in the Cauchy integral formula as a geometric series and using the uniform convergence of the series to integrate term by term. We assume for simplicity that $z_0 = 0$.

Fix z such that $|z| < r$. For $|\zeta| = r$ we have

$$\frac{1}{\zeta - z} = \frac{1}{\zeta} \frac{1}{1 - z/\zeta} = \frac{1}{\zeta} \sum_{k=0}^{\infty} \left(\frac{z}{\zeta}\right)^k = \sum_{k=0}^{\infty} \frac{z^k}{\zeta^{k+1}},$$

where the series converges uniformly for $|\zeta| = r$. Hence

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=r} \left(\sum_{k=0}^{\infty} f(\zeta) \frac{z^k}{\zeta^{k+1}} \right) d\zeta \\ &= \frac{1}{2\pi i} \sum_{k=0}^{\infty} \left(\int_{|\zeta|=r} \frac{f(\zeta)}{\zeta^{k+1}} d\zeta \right) z^k = \sum_{k=0}^{\infty} a_k z^k, \end{aligned}$$

where a_k is given by (4.3) with $z_0 = 0$. Thus $f(z)$ is represented by a power series, which converges whenever $|z| < r$. Since $r < \rho$ is arbitrary, the radius of convergence of the power series satisfies $R \geq \rho$. The formula (4.2) for a_k coincides with formula (3.2), which was obtained by differentiating the power series k times and substituting $z = z_0$. The estimates (4.4) are another version of the Cauchy estimates already derived in Section IV.4. Recall that they are obtained by applying the ML -estimate to the integral in (4.3).

Example. The exponential function e^z has power series $\sum a_k z^k$, where

$$a_k = \frac{1}{k!} \frac{d^k}{dz^k} e^z \Big|_{z=0} = \frac{1}{k!} e^z \Big|_{z=0} = \frac{1}{k!}.$$

Hence

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots.$$

Since e^z is entire, the radius of convergence of the power series is $R = +\infty$. Similarly, the entire functions $\sin z$ and $\cos z$ have power series expansions with infinite radius of convergence, given by

$$\begin{aligned} \sin z &= \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{(2k+1)!} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots, \\ \cos z &= \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k}}{(2k)!} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots. \end{aligned}$$

We state for emphasis two results that are immediate consequences of the theorem. The first is that an analytic function on a disk is completely determined by its value and the values of its derivatives at the center of the disk. This is because the power series representing $f(z)$ is determined by the derivatives of $f(z)$ at the center z_0 via the formula (4.2) for the power series coefficients.

Corollary. Suppose that $f(z)$ and $g(z)$ are analytic for $|z - z_0| < r$. If $f^{(k)}(z_0) = g^{(k)}(z_0)$ for $k \geq 0$, then $f(z) = g(z)$ for $|z - z_0| < r$.

The second corollary provides another method for determining the radius of convergence of a power series. The slogan form of the method is: *The radius of convergence is the distance to the nearest singularity.* The formal statement is as follows.

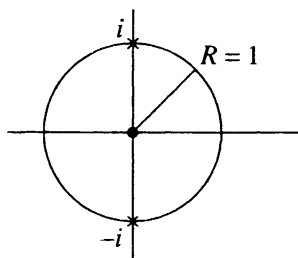
Corollary. Suppose that $f(z)$ is analytic at z_0 , with power series expansion $f(z) = \sum a_k(z - z_0)^k$ centered at z_0 . Then the radius of convergence of the power series is the largest number R such that $f(z)$ extends to be analytic on the disk $\{|z - z_0| < R\}$.

Note that the analytic extension of $f(z)$ to this largest disk is unique, by the first corollary.

Example. The function $1/(1 + x^2)$ is a beautiful function of the real variable x , which is expandable as a real power series about any point on the real axis. Yet the power series about $x = 0$, given by

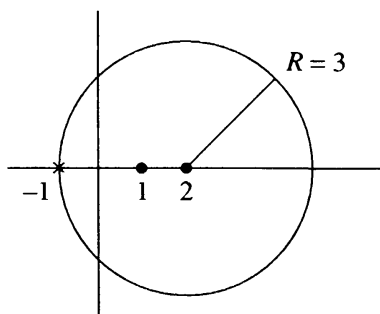
$$\frac{1}{1 + x^2} = 1 - x^2 + x^4 - x^6 + \cdots,$$

has radius of convergence only 1. The reason is that considered as a function $1/(1 + z^2)$ of a complex variable z , the function has singularities at the points $\pm i$. Thus the radius of convergence is necessarily $R = 1$, which is the distance from 0 to the nearest singularities at $\pm i$. This example teaches us that to understand real-valued functions of a real variable, we must look into the complex plane.



singularities at $\pm i$

Example. Consider the power series expansion of the function $f(z) = (z^3 - 1)/(z^2 - 1)$ about $z = 2$, $f(z) = \sum a_k(z - 2)^k$. The function $f(z)$ is analytic in the entire complex plane except for apparent singularities at $z = \pm 1$. However, the singularity at $z = +1$ is illusory. If we eliminate the common factor of $z - 1$ from the numerator and the denominator, we obtain $f(z) = (z^2 + z + 1)/(z + 1)$, which is analytic except at $z = -1$. The apparent singularity at $z = +1$ is called a “removable singularity.” We



singularity at -1 , analytic at $+1$

will return to these in the next chapter. Since $|f(z)| \rightarrow +\infty$ as $z \rightarrow -1$, the function $f(z)$ cannot be extended analytically to $z = -1$, and the singularity at $z = -1$ is genuine. The largest disk centered at $z = 2$ to which $f(z)$ extends analytically is then the disk $\{|z - 2| < 3\}$. Consequently, the radius of convergence of the power series is $R = 3$.

Example. The radius of convergence of the power series $\sum a_k(z - 5)^k$ of the function $(\operatorname{Log} z)/(z - 1)$ about $z = 5$ is $R = 5$. While the function has an apparent singularity at $z = 1$, in fact it extends analytically to that point. Its power series representation about $z = 1$ can be obtained by dividing the power series representation of $\operatorname{Log} z$ by $z - 1$,

$$\frac{\operatorname{Log} z}{z - 1} = 1 - \frac{1}{2}(z - 1) + \frac{1}{3}(z - 1)^2 - \frac{1}{4}(z - 1)^3 + \dots$$

Since the function tends to $+\infty$ as z tends to 0 from along the positive real axis, the function cannot be extended analytically to $z = 0$. Thus $z = 0$ is a genuine singularity of the function, of a type called a “branch point,” and the radius of the largest disk centered at $z = 5$ to which $(\operatorname{Log} z)/(z - 1)$ extends analytically is $R = 5$.

Exercises for V.4

- Find the radius of convergence of the power series for the following functions, expanded about the indicated point.

(a) $\frac{1}{z - 1}$, about $z = i$,	(d) $\operatorname{Log} z$, about $z = 1 + 2i$,
(b) $\frac{1}{\cos z}$, about $z = 0$,	(e) $z^{3/2}$, about $z = 3$,
(c) $\frac{1}{\cosh z}$, about $z = 0$,	(f) $\frac{z - i}{z^3 - z}$, about $z = 2i$.
- Show that the radius of convergence of the power series expansion of $(z^2 - 1)/(z^3 - 1)$ about $z = 2$ is $\sqrt{7}$.
- Find the power series expansion of $\operatorname{Log} z$ about the point $z = i - 2$. Show that the radius of convergence of the series is $R = \sqrt{5}$. Explain why this does not contradict the discontinuity of $\operatorname{Log} z$ at $z = -2$.

4. Suppose $f(z)$ is analytic at $z = 0$ and satisfies $f(z) = z + f(z)^2$. What is the radius of convergence of the power series expansion of $f(z)$ about $z = 0$?
5. Deduce the identity $e^{iz} = \cos z + i \sin z$ from the power series expansions.
6. Find the power series expansions of $\cosh z$ and $\sinh z$ about $z = 0$. What are the radii of convergence of the series?
7. Find the power series expansion of the principal branch $\text{Tan}^{-1}(z)$ of the inverse tangent function about $z = 0$. What is the radius of convergence of the series? *Hint.* Find it by integrating its derivative (a geometric series) term by term.
8. Expand $\text{Log}(1 + iz)$ and $\text{Log}(1 - iz)$ in power series about $z = 0$. By comparing power series expansions (see the preceding exercise), establish the identity

$$\text{Tan}^{-1} z = \frac{1}{2i} \text{Log} \left(\frac{1 + iz}{1 - iz} \right).$$

(See Exercise 5 in Section I.8.)

9. Let a be real, and consider the branch of z^a that is real and positive on $(0, \infty)$. Expand z^a in a power series about $z = 1$. What is the radius of convergence of the series? Write down the series explicitly.
10. Recall that for a complex number α , the binomial coefficient “ α choose n ” is defined by

$$\binom{\alpha}{0} = 1, \quad \text{and} \quad \binom{\alpha}{n} = \frac{\alpha(\alpha - 1) \cdots (\alpha - n + 1)}{n!}, \quad n \geq 1.$$

Find the radius of convergence of the binomial series

$$\sum_{n=0}^{\infty} \binom{\alpha}{n} z^n.$$

Show that the binomial series represents the principal branch of the function $(1 + z)^\alpha$. For which α does the binomial series reduce to a polynomial?

11. For fixed $n \geq 0$, define the function $J_n(z)$ by the power series

$$J_n(z) = \sum_{k=0}^{\infty} \frac{(-1)^k z^{n+2k}}{k!(n+k)! 2^{n+2k}}.$$

Show that $J_n(z)$ is an entire function. Show that $w = J_n(z)$ satisfies the differential equation

$$w'' + \frac{1}{z}w' + \left(1 - \frac{n^2}{z^2}\right)w = 0.$$

Remark. This is **Bessel's differential equation**, and $J_n(z)$ is **Bessel's function** of order n .

12. Suppose that the analytic function $f(z)$ has power series expansion $\sum a_n z^n$. Show that if $f(z)$ is an even function, then $a_n = 0$ for n odd. Show that if $f(z)$ is an odd function, then $a_n = 0$ for n even.
13. Prove the following version of L'Hospital's rule. If $f(z)$ and $g(z)$ are analytic, $f(z_0) = g(z_0) = 0$, and $g(z)$ is not identically zero, then

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \lim_{z \rightarrow z_0} \frac{f'(z)}{g'(z)},$$

in the sense that either both limits are finite and equal, or both limits are infinite.

14. Let f be a continuous function on the unit circle $T = \{|z| = 1\}$. Show that f can be approximated uniformly on T by a sequence of polynomials in z if and only if f has an extension F that is continuous on the closed disk $\{|z| \leq 1\}$ and analytic on the interior $\{|z| < 1\}$. *Hint.* To approximate such an F , consider dilates $F_r(z) = F(rz)$.

5. Power Series Expansion at Infinity

We say that the function $f(z)$ is **analytic at** $z = \infty$ if the function $g(w) = f(1/w)$ is analytic at $w = 0$. Thus we make a change of variable $w = 1/z$, $z = 1/w$, and we study the behavior of $f(z)$ at $z = \infty$ by studying the behavior of $g(w)$ at $w = 0$.

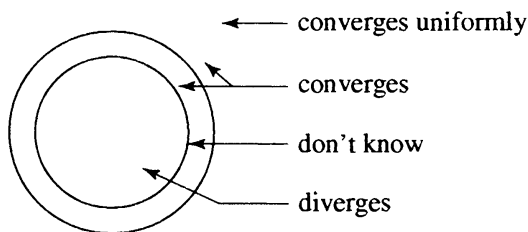
If $f(z)$ is analytic at ∞ , then $g(w) = f(1/w)$ has a power series expansion centered at $w = 0$,

$$g(w) = \sum_{k=0}^{\infty} b_k w^k = b_0 + b_1 w + b_2 w^2 + b_3 w^3 + \cdots, \quad |w| < \rho.$$

Thus $f(z)$ is represented by a convergent series expansion in descending powers of z ,

$$(5.1) \quad f(z) = \sum_{k=0}^{\infty} \frac{b_k}{z^k} = b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3} + \cdots, \quad |z| > \frac{1}{\rho}.$$

This series converges absolutely for $|z| > 1/\rho$, and for any $r > 1/\rho$ it converges uniformly for $|z| \geq r$.



A formula for the coefficients can be obtained by multiplying the series by z^m and integrating term by term around the circle $|z| = r$. We have

$$\begin{aligned} \int_{|z|=r} f(z) z^m dz &= \int_{|z|=r} \left(\sum b_k z^{-k} \right) z^m dz \\ &= \sum b_k \int_{|z|=r} z^{m-k} dz = 2\pi i b_{m+1}. \end{aligned}$$

Thus the coefficient b_k of $1/z^k$ is given by

$$b_k = \frac{1}{2\pi i} \int_{|z|=r} f(z) z^{k-1} dz, \quad k \geq 0.$$

This formula should not be memorized. It will be superseded by the formula for the coefficients of a Laurent expansion, to be derived in Chapter VI.

Example. If $n \geq 0$, the function $f(z) = 1/z^n$ is analytic at ∞ , since $g(w) = w^n$ is analytic at $w = 0$.

Example. Consider the function $f(z) = 1/(z^2 + 1)$. In this case,

$$g(w) = f(1/w) = \frac{1}{(1/w)^2 + 1} = \frac{w^2}{1 + w^2}.$$

Since $g(w)$ is analytic at $w = 0$, $f(z)$ is analytic at ∞ . The power series for $g(w)$ is obtained by expanding $1/(1 + w^2)$ in a geometric series, to obtain

$$g(w) = w^2 \sum_{k=0}^{\infty} (-1)^k w^{2k} = w^2 - w^4 + w^6 - w^8 + \cdots, \quad |w| < 1.$$

Thus

$$f(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{z^{2k}} = \frac{1}{z^2} - \frac{1}{z^4} + \frac{1}{z^6} - \frac{1}{z^8} + \cdots, \quad |z| > 1.$$

This expansion can also be obtained by expressing

$$\frac{1}{1 + z^2} = \frac{1}{z^2} \frac{1}{1 + 1/z^2}$$

and expanding $1/(1 + 1/z^2)$ in a geometric series.

Exercises for V.5

1. Expand the following functions in power series about ∞ :

$$(a) \frac{1}{z^2 + 1} \quad (b) \frac{z^2}{z^3 - 1} \quad (c) e^{1/z^2} \quad (d) z \sinh(1/z)$$

2. Suppose $f(z)$ is analytic at ∞ , with series expansion (5.1). With the notation $f(\infty) = b_0$ and $f'(\infty) = b_1$, show that

$$f'(\infty) = \lim_{z \rightarrow \infty} z[f(z) - f(\infty)].$$

3. Suppose $f(z)$ is analytic at ∞ , with series expansion (5.1). Let $\sigma \geq 0$ be the smallest number such that $f(z)$ extends to be analytic for $|z| > \sigma$. Show that the series (5.1) converges absolutely for $|z| > \sigma$ and diverges for $|z| < \sigma$.
4. Let E be a bounded subset of the complex plane $\mathbb{C}$ over which area integrals can be defined, and set

$$f(w) = \iint_E \frac{dx dy}{w - z}, \quad w \in \mathbb{C} \setminus E,$$

where $z = x + iy$. Show that $f(w)$ is analytic at ∞ , and find a formula for the coefficients of the power series of $f(w)$ at ∞ in descending powers of w . *Hint.* Use a geometric series expansion.

5. Determine explicitly the function $f(w)$ defined in Exercise 4, in the case that $E = \{|w| \leq 1\}$ is the unit disk. *Hint.* There are two formulae for $f(w)$, one valid for $|w| \geq 1$ and the other for $|w| \leq 1$. Be sure they agree for $|w| = 1$.

6. Manipulation of Power Series

Power series are easy to work with. For all practical purposes, power series can be treated like polynomials. We have already seen that power series can be differentiated term by term and that they can be integrated term by term. Power series can also be added and multiplied, just like polynomials.

Suppose, for instance, that $f(z)$ and $g(z)$ are analytic at 0, with power series representations

$$f(z) = \sum_{k=0}^{\infty} a_k z^k, \quad g(z) = \sum_{k=0}^{\infty} b_k z^k.$$

Then the power series of the sum $f(z) + g(z)$ is obtained by simply adding coefficients,

$$f(z) + g(z) = \sum_{k=0}^{\infty} (a_k + b_k) z^k.$$

If c is a complex constant, the power series of $cf(z)$ is obtained by multiplying coefficients by c ,

$$cf(z) = \sum_{k=0}^{\infty} ca_k z^k.$$

The formula for the coefficients of the power series of the product $f(z)g(z)$ is more complicated, though it is the same as for products of polynomials. If

$$f(z)g(z) = \sum_{k=0}^{\infty} c_k z^k,$$

then the coefficients c_k are given by

$$(6.1) \quad c_k = a_k b_0 + a_{k-1} b_1 + \cdots + a_1 b_{k-1} + a_0 b_k, \quad k \geq 0.$$

This can be justified as follows. The partial sums $f_n(z) = \sum_{j=0}^n a_j z^j$ and $g_n(z) = \sum_{k=0}^n a_k z^k$ are sequences of polynomials that converge uniformly to $f(z)$ and $g(z)$, respectively, in some disk centered at 0 as $n \rightarrow \infty$. Consequently, $f_n(z)g_n(z)$ converges uniformly to $f(z)g(z)$ on the disk. It follows (Section 2) that the derivatives of $f_n(z)g_n(z)$ converge to the corresponding derivatives of $f(z)g(z)$, and by formula (4.2), the coefficient of z^k in the polynomial $f_n(z)g_n(z)$ converges to the power series coefficient of z^k for $f(z)g(z)$. Now the coefficients of $f_n(z)g_n(z)$ are obtained by multiplying the polynomials and gathering terms,

$$\begin{aligned} f_n(z)g_n(z) &= \left(\sum_{j=0}^n a_j z^j \right) \left(\sum_{j=0}^n a_j z^j \right) = a_0 b_0 + (a_1 b_0 + a_0 b_1)z + \cdots \\ &\quad \cdots + (a_n b_0 + a_{n-1} b_1 + \cdots + a_0 b_n)z^n + \mathcal{O}(z^{n+1}), \end{aligned}$$

where here the notation $\mathcal{O}(z^m)$ is used for terms involving powers z^k for $k \geq m$. (This is consistent with our earlier use of the “big-oh” notation. See Exercise VI.2.6.) For $k \leq n$ the coefficient of z^k in this polynomial is exactly c_k given by (6.1). Passing to the limit as $n \rightarrow \infty$, we find that $f(z)g(z)$ also has power series coefficient of z^k equal to c_k .

The power series of a quotient $f(z)/g(z)$ can also be effectively computed. It suffices to compute the power series of $1/g(z)$. For this, we assume that $g(z)$ is analytic at $z = 0$, and we suppose for simplicity that $g(0) = 1$. The power series expansion of $g(z)$ then has the form

$$g(z) = 1 + \sum_{k=1}^{\infty} b_k z^k = 1 + b_1 z + b_2 z^2 + \cdots.$$

If z is near 0, the sum $\sum_{k=1}^{\infty} b_k z^k$ is small, and we can expand $1/g(z)$ in a geometric series

$$\begin{aligned}\frac{1}{g(z)} &= \frac{1}{1 + \sum_{k=1}^{\infty} b_k z^k} \\ &= 1 - \left(\sum_{k=1}^{\infty} b_k z^k \right) + \left(\sum_{k=1}^{\infty} b_k z^k \right)^2 - \left(\sum_{k=1}^{\infty} b_k z^k \right)^3 + \cdots.\end{aligned}$$

The terms involving z^m occur only in the first $m+1$ summands. We can compute the coefficients of the power series expansion of $1/g(z)$ by collecting the coefficients of z^m in each of the first $m+1$ summands and discarding the remaining summands. The procedure is justified by the uniform convergence of the geometric series for z near 0.

Example. To find the coefficients of z^m for $m \leq 5$ in the power series expansion of $\tan z = \sin z / \cos z$ about $z = 0$, we calculate as follows, again using the notation $\mathcal{O}(z^m)$ for terms involving powers z^k for $k \geq m$:

$$\begin{aligned}\frac{1}{\cos z} &= \frac{1}{1 - (z^2/2!) + (z^4/4!) + \mathcal{O}(z^6)} \\ &= 1 + \left(\frac{z^2}{2!} - \frac{z^4}{4!} + \mathcal{O}(z^6) \right) + \left(\frac{z^2}{2!} - \frac{z^4}{4!} + \mathcal{O}(z^6) \right)^2 + \mathcal{O}(z^6) \\ &= 1 + \frac{1}{2}z^2 + \frac{5}{24}z^4 + \mathcal{O}(z^6),\end{aligned}$$

so that

$$\begin{aligned}\frac{\sin z}{\cos z} &= \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} + \mathcal{O}(z^7) \right) \left(1 + \frac{1}{2}z^2 + \frac{5}{24}z^4 + \mathcal{O}(z^6) \right) \\ &= z + \frac{1}{3}z^3 + \frac{2}{15}z^5 + \mathcal{O}(z^7).\end{aligned}$$

The end result can be checked by differentiating $\tan z$ five times. Note that $\tan z$ is an odd function, so that only odd terms appear in the power series.

Exercises for V.6

1. Calculate the terms through order seven of the power series expansion about $z = 0$ of the function $1/\cos z$.
2. Calculate the terms through order five of the power series expansion about $z = 0$ of the function $z/\sin z$.
3. Show that

$$\frac{e^z}{1+z} = 1 + \frac{1}{2}z^2 - \frac{1}{3}z^3 + \frac{3}{8}z^4 - \frac{11}{30}z^5 + \cdots.$$

Show that the general term of the power series is given by

$$a_n = (-1)^n \left[\frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} \right], \quad n \geq 2.$$

What is the radius of convergence of the series?

4. Define the **Bernoulli numbers** B_n by

$$\frac{z}{2} \cot(z/2) = 1 - B_1 \frac{z^2}{2!} - B_2 \frac{z^4}{4!} - B_3 \frac{z^6}{6!} - \cdots.$$

Explain why there are no odd terms in this series. What is the radius of convergence of the series? Find the first five Bernoulli numbers.

5. Define the **Euler numbers** E_n by

$$\frac{1}{\cosh z} = \sum_{n=0}^{\infty} \frac{E_n}{n!} z^n.$$

What is the radius of convergence of the series? Show that $E_n = 0$ for n odd. Find the first four nonzero Euler numbers.

6. Show that the coefficients of a power series “depend continuously” on the function they represent, in the following sense. If $\{f_m(z)\}$ is a sequence of analytic functions that converges uniformly to $f(z)$ for $|z| < \rho$, and

$$f_m(z) = \sum_{k=0}^{\infty} a_{k,m} z^k, \quad f(z) = \sum_{k=0}^{\infty} a_k z^k,$$

then for each $k \geq 0$, we have $a_{k,m} \rightarrow a_k$ as $m \rightarrow \infty$.

7. The Zeros of an Analytic Function

Let $f(z)$ be analytic at z_0 , and suppose that $f(z_0) = 0$ but $f(z)$ is not identically zero. We say that $f(z)$ has a **zero of order** N at z_0 if $f(z_0) = f'(z_0) = \cdots = f^{(N-1)}(z_0) = 0$, while $f^{(N)}(z_0) \neq 0$. In view of the formula (4.2) for the power series coefficients, this occurs if and only if the power series expansion of $f(z)$ has the form

$$f(z) = a_N (z - z_0)^N + a_{N+1} (z - z_0)^{N+1} + \cdots,$$

where $a_N \neq 0$. We can factor out the term $(z - z_0)^N$ from the power series and write

$$(7.1) \quad f(z) = (z - z_0)^N h(z),$$

where $h(z)$ is analytic at z_0 and $h(z_0) = a_N \neq 0$. Conversely, if there is a factorization (7.1) where $h(z)$ is analytic at z_0 and $h(z_0) \neq 0$, then the

leading term in the power series for $f(z)$ is $h(z_0)(z - z_0)^N$, and $f(z)$ has a zero of order N at z_0 .

A zero of order one is called a **simple zero**, and a zero of order two is called a **double zero**.

Example. The zeros of $\sin z$ are at the points $n\pi$, $-\infty < n < +\infty$, and the derivative $\cos z$ of $\sin z$ is ± 1 at each of these points. Hence all zeros of $\sin z$ are simple zeros.

Example. The monomial $(z - z_0)^n$ has a zero of order n at z_0 and no other zeros.

Example. From the power series expansion

$$\sin z = -(z - \pi) + \frac{1}{3!}(z - \pi)^3 - \frac{1}{5!}(z - \pi)^5 + \cdots,$$

we see that the function $\sin z + z - \pi$ has a triple zero at $z = \pi$. The function $\frac{\sin z}{z - \pi} + 1$, defined to be 0 at $z = \pi$, is entire and has a double zero at $z = \pi$, since the leading term of its power series expansion is $(z - \pi)^2/3!$.

A useful rule for determining orders of zeros is that the order of a zero of a product $f(z)g(z)$ is the sum of the orders of the corresponding zeros of the factors $f(z)$ and $g(z)$. Indeed, if $f(z) = a_n(z - z_0)^n + \cdots$ has a zero of order n at z_0 and $g(z) = b_m(z - z_0)^m + \cdots$ has a zero of order m at z_0 , then $f(z)g(z) = a_nb_m(z - z_0)^{n+m} + \cdots$ has a zero of order $m + n$ at z_0 .

If $f(z)$ is analytic at ∞ and $f(\infty) = 0$, we define the order of the zero of $f(z)$ at $z = \infty$ in the usual way, by making the change of variable $w = 1/z$. We say that $f(z)$ has a **zero at $z = \infty$ of order N** if $g(w) = f(1/w)$ has a zero at $w = 0$ of order N . In this case, $g(w) = b_N w^N + b_{N+1} w^{N+1} + \cdots$, where $b_N \neq 0$. Thus $f(z)$ has the series representation

$$f(z) = \frac{b_N}{z^N} + \frac{b_{N+1}}{z^{N+1}} + \cdots, \quad |z| > R,$$

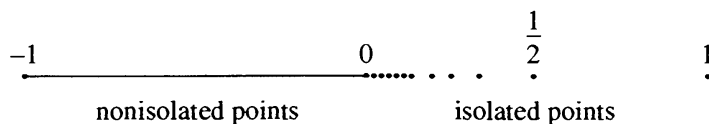
where $b_N \neq 0$.

Example. The function $1/(1 + z^2)$ has a double zero at ∞ . Its power series expansion, derived in Section 5, is $1/z^2 - 1/z^4 + \cdots$, which has leading term $1/z^2$.

Example. The rational monomial $1/(z - z_0)^n$ has a zero of order n at ∞ .

We say that a point $z_0 \in E$ is an **isolated point** of the set E if there is $\rho > 0$ such that $|z - z_0| \geq \rho$ for all points $z \in E$ other than z_0 . In other words, z_0 is an isolated point of E if z_0 is at a positive distance from $E \setminus \{z_0\}$. If E is a set such that each point of E is an isolated point of E , we say that the “points of E are isolated.”

Example. If $E = [-1, 0] \cup \{1/n : n \geq 1\}$, then each of the points $1/n$ is an isolated point of E , while no point of the interval $[-1, 0]$ is an isolated point of E .



Theorem. If D is a domain, and $f(z)$ is an analytic function on D that is not identically zero, then the zeros of $f(z)$ are isolated.

The proof of this theorem breaks into two parts, an observation about the local behavior of an analytic function near a zero, and a connectedness argument that depends on D being a domain. We begin with the connectedness argument, which is deceptively subtle.

Let U be the set of all $z \in D$ such that $f^{(m)}(z) = 0$ for all $m \geq 0$. If $z_0 \in U$, then the power series expansion $f(z) = \sum a_k(z - z_0)^k$ has $a_k = f^{(k)}(z_0)/k! = 0$ for all $k \geq 0$. Hence $f(z) = 0$ for z belonging to a disk centered at z_0 . The points of this disk all belong to U . This shows that U is an open set. On the other hand, if $z_0 \in D \setminus U$, then $f^{(k)}(z_0) \neq 0$ for some k . Therefore, $f^{(k)}(z) \neq 0$ for z in some disk centered at z_0 , and this disk is contained in $D \setminus U$, so $D \setminus U$ is also open. Since D is connected, either $U = D$ or U is empty. If $U = D$, then $f(z) = 0$ for all $z \in D$, contrary to our hypothesis. Hence U is empty. Thus we conclude from the connectedness argument that each zero of $f(z)$ has finite order.

The closing argument is easier. If z_0 is a zero of $f(z)$, say of order N , we can factor $f(z) = (z - z_0)^N h(z)$, where $h(z)$ is analytic at z_0 and $h(z_0) \neq 0$. Then for $\rho > 0$ sufficiently small, we have $h(z) \neq 0$ for $|z - z_0| < \rho$, and consequently $|f(z)| \neq 0$ for $0 < |z - z_0| < \rho$. Thus $f(z)$ has distance at least ρ from any other zero of $f(z)$, and the zeros of $f(z)$ are isolated.

By applying the preceding theorem to $f(z) - g(z)$, we obtain immediately the following important result, which is also referred to as the **identity principle**.

Theorem (Uniqueness Principle). If $f(z)$ and $g(z)$ are analytic on a domain D , and if $f(z) = g(z)$ for z belonging to a set that has a nonisolated point, then $f(z) = g(z)$ for all $z \in D$.

Example. Once we know that $\sin z$ and $\cos z$ are entire functions that satisfy $\sin^2 x + \cos^2 x = 1$ for all real numbers x , then necessarily $\sin^2 z + \cos^2 z = 1$ for all complex numbers z . This follows from the uniqueness principle, applied to $f(z) = \sin^2 z + \cos^2 z$ and $g(z) = 1$.

The uniqueness principle has a natural extension to functions of two complex variables, which is sometimes referred to as the **principle of permanence of functional equations**.

Theorem. Let D be a domain, and let E be a subset of D that has a nonisolated point. Let $F(z, w)$ be a function defined for $z, w \in D$ such that $F(z, w)$ is analytic in z for each fixed $w \in D$ and analytic in w for each fixed $z \in D$. If $F(z, w) = 0$ whenever z and w both belong to E , then $F(z, w) = 0$ for all $z, w \in D$.

This follows from two applications of the uniqueness principle, one for each variable. First fix $z_0 \in E$. Then $F(z_0, w)$ is analytic for $w \in D$ and vanishes for $w \in E$. By the uniqueness principle, $F(z_0, w) = 0$ for all $w \in D$. Now fix $w \in D$. We have shown that $F(z_0, w) = 0$ for all $z_0 \in E$. By the uniqueness principle, $F(z, w) = 0$ for all $z \in D$.

Example. As a typical application, we derive the addition formula for the exponential function, assuming that $e^{s+t} = e^s e^t$ for s and t real. The function $F(z, w) = e^{z+w} - e^z e^w$ is an entire function of each variable for fixed values of the other variable, and it vanishes when both the variables are real. By the permanence principle, it then vanishes for all values of z and w . Thus $e^{z+w} = e^z e^w$.

Exercises for V.7

- Find the zeros and orders of zeros of the following functions.

(a) $\frac{z^2 + 1}{z^2 - 1}$	(d) $\cos z - 1$	(g) $e^z - 1$
(b) $\frac{1}{z} + \frac{1}{z^5}$	(e) $\frac{\cos z - 1}{z}$	(h) $\sinh^2 z + \cosh^2 z$
(c) $z^2 \sin z$	(f) $\frac{\cos z - 1}{z^2}$	(i) $\frac{\operatorname{Log} z}{z}$ (principal value)
- Determine which of the functions in the preceding exercise are analytic at ∞ , and determine the orders of any zeros at ∞ .
- Show that the zeros of $\sin z$ and $\tan z$ are all simple.
- Show that $\cos(z + w) = \cos z \cos w - \sin z \sin w$, assuming the corresponding identity for z and w real.
- Show that

$$\int_{-\infty}^{\infty} e^{-zt^2 + 2wt} dt = \sqrt{\frac{\pi}{z}} e^{w^2/z}, \quad z, w \in \mathbb{C}, \operatorname{Re} z > 0,$$

where we take the principal branch of the square root. Compare the result to Exercise IV.3.1. *Hint.* Show that the integral is analytic

in z and w , and evaluate it for $z = x > 0$ and w real by making a change of variable and using the known value $\sqrt{\pi}$ for $z = 1$ and $w = 0$.

6. Suppose $f(z)$ is analytic on a domain D and $z_0 \in D$. Show that if $f^{(m)}(z_0) = 0$ for $m \geq 1$, then $f(z)$ is constant on D .
7. Show that if $u(x, y)$ is a harmonic function on a domain D such that all the partial derivatives of $u(x, y)$ vanish at the same point of D , then $u(x, y)$ is constant on D .
8. With the convention that the function that is identically zero has a zero of infinite order at each point, show that if $f(z)$ and $g(z)$ have zeros of order n and m respectively at z_0 , then $f(z) + g(z)$ has a zero of order $k \geq \min(n, m)$. Show that strict inequality can occur here, but that equality holds whenever $m \neq n$.
9. Show that if the analytic function $f(z)$ has a zero of order N at z_0 , then $f(z) = g(z)^N$ for some function $g(z)$ analytic near z_0 and satisfying $g'(z_0) \neq 0$.
10. Show that if $f(z)$ is a continuous function on a domain D such that $f(z)^N$ is analytic on D for some integer N , then $f(z)$ is analytic on D .
11. Show that if $f(z)$ is a nonconstant analytic function on a domain D , then the image under $f(z)$ of any open set is open. *Remark.* This is the **open mapping theorem** for analytic functions. The proof is easy when $f'(z) \neq 0$, since the Jacobian of $f(z)$ coincides with $|f'(z)|^2$. Use Exercise 9 to deal with the points where $f'(z)$ is zero.
12. Show that the open mapping theorem for analytic functions implies the maximum principle for analytic functions.
13. Let $f_n(z)$ be a sequence of analytic functions on a domain D such that $f_n(D) \subset D$, and suppose that $f_n(z)$ converges normally to $f(z)$ on D . Show that either $f(D) \subset D$, or else $f(D)$ consists of a single point in ∂D .
14. A set E is **discrete** if every point of E is isolated. Show that a closed discrete subset of a domain D either is finite or can be arranged in a sequence $\{z_k\}$ that accumulates only on $\{\infty\} \cup \partial D$.

8. Analytic Continuation

In Chapter I we analyzed the branches of the functions $\sqrt{z}$ and $\log z$ by following the values of the functions along curves in the complex plane. By

this method we saw that these functions cannot be extended continuously to $\mathbb{C} \setminus \{0\}$. We constructed Riemann surfaces to which the functions *do* extend continuously, by defining branches of the functions on separate sheets and pasting these sheets together.

In this section we develop more formally the idea of tracking an analytic function along a path. The power series expansion of an analytic function about a point contains complete information about the function near the point. Rather than track only the values of the analytic function, we will track the power series expansions of the function about points of the path as we move along the path. We begin by observing that the radius of convergence of a power series depends continuously on the center of the expansion.

Lemma. *Suppose D is a disk, $f(z)$ is analytic on D , and $R(z_1)$ is the radius of convergence of the power series expansion of $f(z)$ about a point $z_1 \in D$. Then*

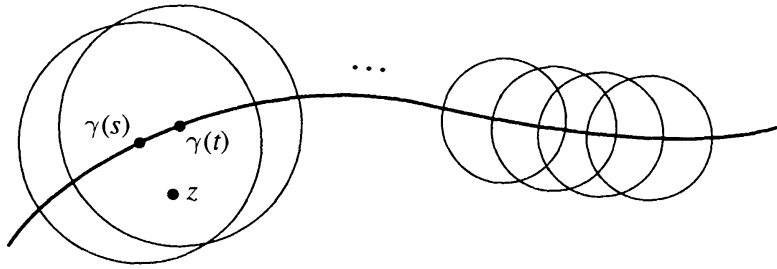
$$(8.1) \quad |R(z_1) - R(z_2)| \leq |z_1 - z_2|, \quad z_1, z_2 \in D.$$

We use the characterization of $R(z_1)$ as the radius of the largest disk centered at z_1 to which $f(z)$ extends analytically. Thus $f(z)$ does not extend analytically to any disk containing $\{|z - z_1| \leq R(z_1)\}$, and consequently $R(z_2) \leq R(z_1) + |z_2 - z_1|$. Interchanging z_1 and z_2 , we obtain also $R(z_1) \leq R(z_2) + |z_2 - z_1|$. These two inequalities yield (8.1).

Now we start with a power series $\sum a_n(z - z_0)^n$ that represents a function $f(z)$ near z_0 . We are interested in the behavior of $f(z)$ only near z_0 , and we say that the power series represents the “germ” of $f(z)$ at z_0 . Let $\gamma(t)$, $a \leq t \leq b$, be a path starting at $z_0 = \gamma(a)$. We say that $f(z)$ is **analytically continuable along** γ if for each t there is a convergent power series

$$(8.2) \quad f_t(z) = \sum_{n=0}^{\infty} a_n(t)(z - \gamma(t))^n, \quad |z - \gamma(t)| < r(t),$$

such that $f_a(z)$ is the power series representing $f(z)$ at z_0 , and such that when s is near t , then $f_s(z) = f_t(z)$ for z in the intersection of the disks of convergence. By the uniqueness principle, the series $f_t(z)$ determines uniquely each of the series $f_s(z)$ for s near t . It follows that the series $f_b(z)$ is uniquely determined by $f_a(z)$. (Otherwise, we could define t_0 to be the infimum of the parameter values t such that $f_t(z)$ is *not* uniquely determined by $f_a(z)$, and we would soon have a contradiction to the local uniqueness assertion at t_0 .) We refer to $f_b(z)$ as the **analytic continuation** of $f(z)$ along γ , where we regard $f_b(z)$ either as a power series or as an analytic function defined near $\gamma(b)$. Since the coefficients $a_n(t)$ in (8.2) are given by $a_n(s) = f_t^{(m)}(\gamma(s))/m!$ for s near t , the coefficients depend continuously on the parameter t . The preceding lemma shows that the



radius of convergence of the power series (8.2) also depends continuously on the parameter t . We summarize and give some examples.

Theorem. Suppose $f(z)$ can be continued analytically along the path $\gamma(t)$, $a \leq t \leq b$. Then the analytic continuation is unique. Further, for each $n \geq 0$ the coefficient $a_n(t)$ of the series (8.2) depends continuously on t , and the radius of convergence of the series (8.2) depends continuously on t .

Example. Suppose $f(z)$ is analytic in a domain D . Then $f(z)$ has an analytic continuation along any path in D . Simply define $f_t(z)$ to be the power series expansion of $f(z)$ about $\gamma(t)$.

Example. Suppose $f(z)$ is the principal branch of $\sqrt{z}$, with series expansion

$$f(z) = 1 + \frac{1}{2}(z-1) - \frac{1}{8}(z-1)^2 + \cdots$$

about $z = 1$. Let $\gamma(t) = e^{it}$, $0 \leq t \leq 2\pi$, be the closed path around the unit circle starting at $\gamma(0) = 1$. Then $f(z)$ has an analytic continuation along γ , and the power series $f_t(z)$ in (8.2) is given explicitly by

$$f_t(z) = e^{it/2} + \frac{e^{-it/2}}{2}(z - e^{it}) - \frac{e^{-3it/2}}{8}(z - e^{it})^2 + \cdots.$$

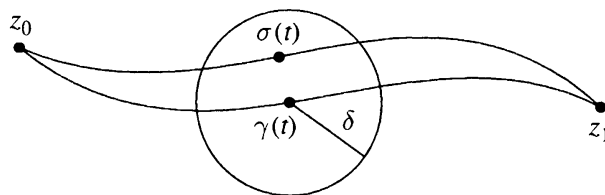
Thus the analytic continuation of $f(z)$ around the circle is

$$f_{2\pi}(z) = -1 - \frac{1}{2}(z-1) + \frac{1}{8}(z-1)^2 + \cdots,$$

which is just the other branch of $\sqrt{z}$.

Now suppose $f(z)$ is analytic at z_0 , and suppose that $\gamma(t)$, $a \leq t \leq b$, is a path from $z_0 = \gamma(a)$ to $z_1 = \gamma(b)$ along which $f(z)$ has an analytic continuation $f_t(z)$. The radius of convergence $R(t)$ of the power series (8.2) varies continuously with t . Hence there is $\delta > 0$ such that $R(t) \geq \delta$ for all t , $a \leq t \leq b$.

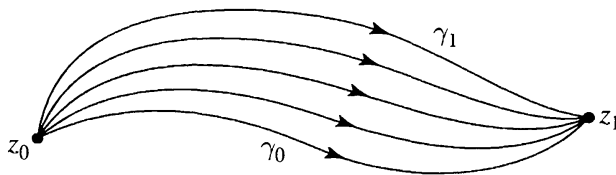
Lemma. Let f , γ , and δ be as above. If $\sigma(t)$, $a \leq t \leq b$, is another path from z_0 to z_1 such that $|\sigma(t) - \gamma(t)| < \delta$ for $a \leq t \leq b$, then there is an analytic continuation $g_t(z)$ of $f_t(z)$ along σ , and the terminal series $g_b(z)$ centered at $\sigma(b) = z_1$ coincides with $f_b(z)$.



We can define $g_t(z)$ to be the power series expansion of $f_t(z)$ about $\sigma(t)$. This is possible, since $\sigma(t)$ lies within the disk of convergence of $f_t(z)$. In fact, $\sigma(s)$ lies inside the disk of convergence of $f_t(z)$ for s near t , and since $f_s(z) = f_t(z)$ for s near t , we see that $g_s(z) = g_t(z)$ for s near t , and so $g_t(z)$ does represent an analytic continuation. By its definition, $g_b(z)$ coincides with $f_b(z)$ at $z_1 = \sigma(b) = \gamma(b)$.

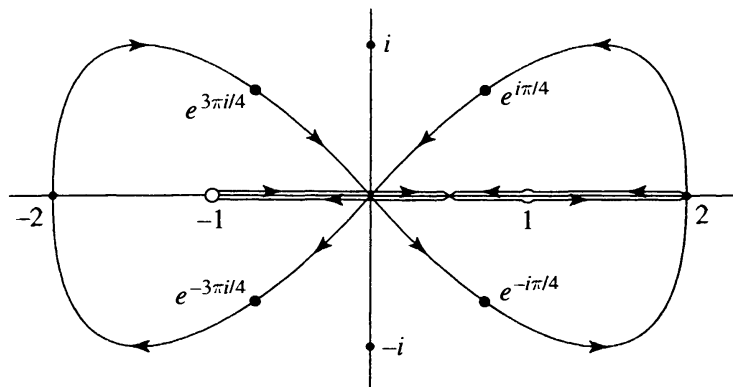
Just as in Section III.2, we can deduce from this “local” deformation result a global deformation theorem.

Theorem (Monodromy Theorem). *Let $f(z)$ be analytic at z_0 . Let $\gamma_0(t)$ and $\gamma_1(t)$, $a \leq t \leq b$, be two paths from z_0 to z_1 along which $f(z)$ can be continued analytically. Suppose $\gamma_0(t)$ can be deformed continuously to $\gamma_1(t)$ by paths $\gamma_s(t)$, $0 \leq s \leq 1$, from z_0 to z_1 such that $f(z)$ can be continued analytically along each path γ_s . Then the analytic continuations of $f(z)$ along γ_0 and along γ_1 coincide at z_1 .*



By a continuous deformation, we mean that the function $(s, t) \mapsto \gamma_s(t)$ is continuous for $0 \leq s \leq 1$ and $a \leq t \leq b$. If $f_{s,t}(z)$ denotes the analytic continuation of $f(z)$ along γ_s , the power series $f_{s,t}(z)$ and $f_{s',t'}(z)$ determine the same function near $\gamma_s(t)$ for s' near s and t' near t . Thus the radius of convergence $R_{s,t}$ of $f_{s,t}$ varies continuously with s and t , and there is $\delta > 0$ such that $R_{s,t} \geq \delta$ for all s and t . We can choose, then, parameter values $0 = s_0 < s_1 < \cdots < s_n = 1$ such that $|\gamma_{s_j}(t) - \gamma_{s_{j-1}}(t)| < \delta$ for $a \leq t \leq b$. By the preceding lemma, the analytic continuation along $\gamma_{s_{j-1}}$ leads to the same power series at z_1 as that along γ_{s_j} . If we apply this now successively to $\gamma_0 = \gamma_{s_0}, \gamma_{s_1}, \gamma_{s_2}, \dots$, we conclude after n steps that analytic continuation along γ_0 leads to the same power series as analytic continuation along γ_1 . This proves the monodromy theorem.

Exercise. Suppose that the principal branch $w = f(z)$ of the algebraic function $(z^4 - 1)^{1/3}$ is continued analytically from $z = 2$ around the figure-eight path indicated below. What is the analytic continuation of the function at the end of the path?



Solution. We deform the figure-eight path to a path for which it is possible to follow the analytic continuation more easily. By the monodromy theorem, the analytic continuation is the same if we follow a path along the real axis except for indentations at ± 1 , as in the figure. For the various segments of this path, we track the behavior of w as follows:

- path from $+2$ to $+1 \rightsquigarrow \arg w = 0$,
- semicircle around $+1 \rightsquigarrow$ phase change of $e^{i\pi/3}$,
- path from $+1$ to $-1 \rightsquigarrow \arg w = e^{i\pi/3}$,
- circle around $-1 \rightsquigarrow$ phase change of $e^{-2\pi i/3}$,
- path from -1 to $+1 \rightsquigarrow \arg w = e^{-i\pi/3}$,
- semicircle around $+1 \rightsquigarrow$ phase change of $e^{i\pi/3}$,
- path from $+1$ to $+2 \rightsquigarrow \arg w = 0$.

Thus at the end of the path we return to the same branch $w = f(z)$ that we started with. This can be checked by factoring $z^4 - 1$ and tracking separately each factor.

Exercises for V.8

- Suppose that the principal branch of $\sqrt{z^2 - 1}$ is continued analytically from $z = 2$ around the figure-eight path indicated above. What is the analytic continuation of the function at the end of the path? Answer the same question for the functions $(z^3 - 1)^{1/3}$ and $(z^6 - 1)^{1/3}$.
- Show that $f(z) = \text{Log } z = (z - 1) - \frac{1}{2}(z - 1)^2 + \cdots$ has an analytic continuation around the unit circle $\gamma(t) = e^{it}$, $0 \leq t \leq 2\pi$. Determine explicitly the power series f_t for each t . How is $f_{2\pi}$ related to f_0 ?
- Show that each branch of $\sqrt{z}$ can be continued analytically along any path γ in $\mathbb{C} \setminus \{0\}$, and show that the radius of convergence of the power series $f_t(z)$ representing the continuation is $|\gamma(t)|$. Show that $\sqrt{z}$ cannot be continued analytically along any path containing 0.

4. Let $f(z)$ be analytic on a domain D , fix $z_0 \in D$, and let $f(z) = \sum a_n(z - z_0)^n$ be the expansion of $f(z)$ about z_0 . Let

$$F(z) = \int_{z_0}^z f(\zeta) d\zeta = \sum_{n=0}^{\infty} \frac{a_n}{n+1} (z - z_0)^{n+1}$$

be the indefinite integral of $f(z)$ for z near z_0 . Show that $F(z)$ can be continued analytically along any path in D starting at z_0 . What happens in the case $D = \mathbb{C} \setminus \{0\}$, $z_0 = 1$, and $f(z) = 1/z$? What happens in the case that D is star-shaped?

5. Show that the function defined by

$$f(z) = \sum z^{2^n} = z + z^2 + z^4 + z^8 + \cdots$$

is analytic on the open unit disk $\{|z| < 1\}$, and that it cannot be extended analytically to any larger open set. *Hint.* Observe that $f(z) = z + f(z^2)$, and that $f(r) \rightarrow +\infty$ as $r \rightarrow 1$.

6. Suppose $f(z) = \sum a_n z^n$, where $a_n = 0$ except for n in a sequence n_k that satisfies $n_{k+1}/n_k \geq 1 + \delta$ for some $\delta > 0$. Suppose further that the series has radius of convergence $R = 1$. Show that $f(z)$ does not extend analytically to any point of the unit circle. *Remark.* Such a sequence with large gaps between successive nonzero terms is called a **lacunary sequence**. This result is the **Hadamard gap theorem**. There is a slick proof. If $f(z)$ extends analytically across $z = 1$, consider $g(w) = f(w^m(1+w)/2)$, where m is a large integer. Show that the power series for $g(w)$ has radius of convergence $r > 1$, and that this implies that the power series of $f(z)$ converges for $1 \leq \operatorname{Re} z < 1 + \varepsilon$.
7. Suppose $f(z) = \sum a_n z^n$, where the series has radius of convergence $R < \infty$. Show that there is an angle α such that $f(z)$ does not have an analytic continuation along the path $\gamma(t) = te^{i\alpha}$, $0 \leq t \leq R$. Determine the radius of convergence of the power series expansion of $f(z)$ about $te^{i\alpha}$.
8. Let $f(z)$ be analytic at z_0 , and let $\gamma(t)$, $a \leq t \leq b$, be a path such that $\gamma(a) = z_0$. If $f(z)$ cannot be continued analytically along γ , show that there is a parameter value t_1 such that there is an analytic continuation $f_t(z)$ for $a \leq t < t_1$, and the radius of convergence of the power series $f_t(z)$ tends to 0 as $t \rightarrow t_1$.
9. Let $P(z, w)$ be a polynomial in z and w , of degree n in w . Suppose that $f(z)$ is analytic at z_0 and satisfies $P(z, f(z)) = 0$. Show that if $f_t(z)$ is any analytic continuation of $f(z)$ along any path starting at z_0 , then $P(z, f_t(z)) = 0$ for all t . *Remark.* An analytic function $f(z)$ that satisfies a polynomial equation $P(z, f(z)) = 0$ is called an

algebraic function. For instance, the branches of $\sqrt[n]{z}$ are algebraic functions, since they satisfy $z - w^n = 0$.

10. Let D be the punctured disk $\{0 < |z| < \varepsilon\}$, suppose $f(z)$ is analytic at $z_0 \in D$, and $e^{w_0} = z_0$. Show that $f(z)$ has an analytic continuation along any path in D starting at z_0 if and only if there is an analytic function $g(w)$ in the half-plane $\{\operatorname{Re} w < \log \varepsilon\}$ such that $f(e^w) = g(w)$ for w near w_0 . *Remark.* If $f(z)$ does not extend analytically to D but has an analytic continuation along any path in D , we say that $f(z)$ has a **branch point** at $z = 0$. For the proof, use the fact that any path in D starting at z_0 is the composition of a unique path in the half-plane starting at w_0 and the exponential function e^w .

VI

Laurent Series and Isolated Singularities

In Section 1 we derive the Laurent decomposition of a function that is analytic on an annulus, and in Section 2 we use the Laurent decomposition on a punctured disk to study isolated singularities of analytic functions. We classify these as removable singularities, essential singularities, or poles, and we characterize each type of singularity. In Section 3 we define isolated singularities at ∞ , and in Section 4 we derive the partial fractions decomposition of a rational function. In Sections 5 and 6 we use the Laurent decomposition to study periodic functions and we relate Laurent series to Fourier series. Sections 5 and 6 can be omitted at first reading.

1. The Laurent Decomposition

The Laurent decomposition splits a function analytic in an annulus as the sum of a function analytic inside the annulus and a function analytic outside the annulus.

Theorem (Laurent Decomposition). *Suppose $0 \leq \rho < \sigma \leq +\infty$, and suppose $f(z)$ is analytic for $\rho < |z - z_0| < \sigma$. Then $f(z)$ can be decomposed as a sum*

$$(1.1) \quad f(z) = f_0(z) + f_1(z),$$

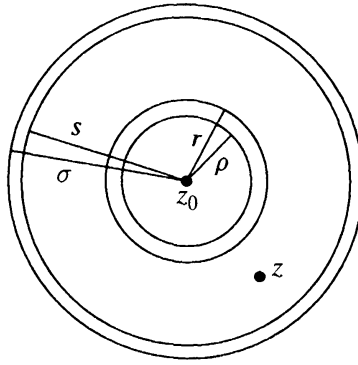
where $f_0(z)$ is analytic for $|z - z_0| < \sigma$, and $f_1(z)$ is analytic for $|z - z_0| > \rho$ and at ∞ . If we normalize the decomposition so that $f_1(\infty) = 0$, then the decomposition is unique.

If $f(z)$ is already analytic for $|z - z_0| < \sigma$, the Laurent decomposition becomes the trivial decomposition $f(z) = f_0(z)$, with $f_1(z) = 0$. If $f(z)$ is already analytic for $|z - z_0| > \rho$ and vanishes at ∞ , the Laurent decomposition is the trivial decomposition $f(z) = f_1(z)$, with $f_0(z) = 0$.

The uniqueness of the decomposition follows from Liouville's theorem by the following argument. Suppose that $f(z) = g_0(z) + g_1(z)$ is another decomposition with the properties of the theorem. Then

$$(1.2) \quad g_0(z) - f_0(z) = f_1(z) - g_1(z), \quad \rho < |z - z_0| < \sigma.$$

Define $h(z)$ to be equal to $g_0(z) - f_0(z)$ in the disk $\{|z - z_0| < \sigma\}$, and equal to $f_1(z) - g_1(z)$ in the exterior domain $\{|z - z_0| > \rho\}$. These domains overlap in the annulus $\{\rho < |z - z_0| < \sigma\}$, and the identity (1.2) shows that the two definitions agree in the overlap. Thus $h(z)$ is defined for all $z \in \mathbb{C}$. Evidently, $h(z)$ is an entire function, and $h(z)$ tends to 0 as $z \rightarrow \infty$. By Liouville's theorem, $h(z)$ is identically zero. Consequently $g_0(z) = f_0(z)$ and $g_1(z) = f_1(z)$, and there is at most one such decomposition.



To find such a decomposition, we apply the Cauchy integral representation theorem on an annulus, as follows. Choose r and s such that $\rho < r < s < \sigma$. The Cauchy integral formula for an annulus yields

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = s} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} d\zeta,$$

which is valid for $r < |z - z_0| < s$. The function

$$f_0(z) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = s} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad |z - z_0| < s,$$

is analytic for $|z - z_0| < s$, and the function

$$f_1(z) = -\frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad |z - z_0| > r,$$

is analytic for $|z - z_0| > r$ and tends to 0 as $z \rightarrow \infty$. Thus we obtain the decomposition $f(z) = f_0(z) + f_1(z)$ for $r < |z - z_0| < s$. Technically, this decomposition depends on r and s . However, the uniqueness assertion already established shows that the decomposition is independent of r and s , so that $f_0(z)$ and $f_1(z)$ are defined for $\rho < |z - z_0| < \sigma$ and have the properties of the theorem.